

Personalize the Corruption, Not the Brain: Reliability-Gated Calibration-Guided Posterior Diffusion for Ocular Artifact Removal in Unseen-User EEG

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Ocular propagation varies across users, sessions, montages, and electrode states, although it may remain locally stable within a validated calibration context. Population cleaners ignore this deployment context, whereas direct subspace subtraction deletes neural activity whenever it overlaps an ocular span. We introduce calibration-guided subspace posterior diffusion (CSPD), a corruption-only personalization framework for unseen-user electroencephalography (EEG). A query-disjoint support recording estimates a context-specific transfer from specified ocular reference regressors to EEG sensors and extracts its leading spatial subspace. At inference, a dataset- and montage-specific clean-EEG diffusion prior is coupled to two observation energies that condition on the same query: a training-only population energy and a support-calibrated context energy. A reliability gate interpolates these energies; at zero reliability, an explicit short-circuit gives the same seeded numerical trajectory as the population-posterior procedure rather than reverting to an unconditional generator. Within the context energy, a temporal attenuation variable scales precision in the ocular span; it is not interpreted as an artifact probability. The resulting finite-variance constraint penalizes disagreement with the observed reliable complement while allowing model-based restoration samples to represent ambiguity in overlapping directions. We derive the complement and subtraction identities, finite estimated-projector bounds, a top-rank subspace perturbation result, the exact fixed-energy diffusion score identity, the plug-in approximation used in computation, and pathwise rollback to the population posterior. A staged held-out-user protocol first tests the oracle mechanism, then operator specificity, diffusion versus both information-matched and strongly supervised one-step estimators, physiological preservation, and learned reliability under drift. Robust, lagged, frequency-selective, graph, covariance-contrast, and population-shrunk operators are retained as a deferred contingency portfolio and are activated only after a diagnosed failure of the primary model. Across [TBD: held-out users and sessions], CSPD changes [TBD: primary paired fidelity metric] by [TBD: estimate and 95% CI] relative to [TBD: frozen comparator] while satisfying [TBD: preservation and rollback gates]. These values remain unset until the clean-room experiments are complete.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → *Health informatics*; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

Additional Key Words and Phrases: electroencephalography, ocular artifacts, adaptive systems, calibration, diffusion models, inverse problems, predictive uncertainty, rollback

1 Introduction

Electroencephalography (EEG) measures voltage differences at the scalp on the scale of microvolts. Blinks, saccades, and eyelid motion generate potentials that are often larger than the neural activity of interest and spread across multiple electrodes. Their effect is consequently neither independent sensor noise nor a fixed waveform that can be removed once and reused. Ocular propagation is not stationary across users, sessions, montages, references, and electrode states, although it can be locally stable within a validated calibration context. An ocular cleaner that works for an average training participant can therefore be systematically wrong for a new user even when the underlying neural phenomenon and task have not changed. This is an adaptive-system problem: the corruption mechanism changes at deployment, the available evidence is limited, and an unreliable update may be worse than refusing context adaptation.

That perspective changes what should be personalized. Subject-adaptive EEG systems often learn a representation of the person or update a large denoising network with target data. Such adaptation entangles neural differences with

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nuisance differences and is difficult to validate from a short, unlabeled recording. We instead advance a narrower principle: *personalize the corruption, not the brain*. A population model supplies a shared prior over clean multichannel EEG. A brief, explicitly separated calibration recording is used only to estimate how ocular sources propagate into the current montage. No subject token, target-specific neural embedding, or query-derived network update is required.

Classical ocular correction already exploits reference channels, component structure, or calibration. Regression is transparent and inexpensive, but a forward regression can remove genuine frontal neural activity correlated with eye movement, whereas backward regression can leave residual contamination when the reference is noisy. Independent component analysis (ICA) and automated component labeling relax the need for explicit references but rely on separability, sufficient data, and stable component identification. Artifact Subspace Reconstruction (ASR) detects high-variance directions relative to a calibration distribution and deterministically reconstructs them from retained covariance directions; it is neither ocular-specific nor a probabilistic posterior. Calibration-based methods such as SGEYESUB identify ocular transfer more directly and apply their own sequential correction matrices. Our oracle-span orthogonal subtraction control isolates the sharper geometric ambiguity: setting an estimated ocular-aligned coordinate to zero deletes any neural component occupying the same sensor span.

Learned denoisers replace hand-designed separation with a population mapping from contaminated to clean signals. Convolutional, recurrent, attention-based, and compact architectures have improved paired waveform metrics on semi-synthetic benchmarks. Task-oriented objectives additionally discourage the cleaner from erasing decoder-relevant structure. Nevertheless, a deterministic population mapping must average over unobserved corruption operators, and a model conditioned on the current observation alone has no guarantee that it can identify which spatial mixing applies to an unseen recording. Diffusion models offer a distributional clean-signal prior and have recently been applied to EEG restoration. Diffusion by itself, however, does not resolve the deployment mismatch. The scientific question is not whether a stochastic model is fashionable, but whether a context-specific observation model and posterior representation recover information that a receptive-field- and parameter-matched conditional estimator cannot.

This paper introduces calibration-guided subspace posterior diffusion (CSPD). During support, synchronized EEG and specified ocular reference regressors identify a context-specific transfer matrix; its training-frozen leading rank defines an exact orthogonal projector. During a disjoint query, the same observed EEG enters both a training-only population observation energy and a context-calibrated energy. The latter applies full finite precision in the estimated complement and a time-varying reliability attenuation in the ocular span. A gate interpolates the two energies, rather than interpolating between an unconditional prior and a calibrated model. In geometric terms, ordinary projection commits the neural component in the ocular span to zero. CSPD instead penalizes disagreement with the measured complement and uses a dataset-specific population prior to infer the remaining ambiguity. This is a soft, finite-variance constraint, not exact equality to the observation.

Calibration is useful only when it is representative. CSPD therefore makes adaptation conditional on auditable evidence. Bootstrap subspace stability, regressor excitation, numerical conditioning, and support–query shift feed a reliability gate learned on training contexts. The gate scales the likelihood-ratio correction from the population observation model to the context model and has a hard refusal state. With shared randomness and gate value zero, the implementation short-circuits every context-dependent operation and follows exactly the same numerical path as a direct population-posterior invocation conditioned on the same query. This is a pathwise software guarantee, not a claim that the finite-step plug-in sampler draws exactly from the ideal posterior. A monitor can lower the gate or request recalibration when drift invalidates the support. This loop—sense, estimate, validate, adapt, monitor, and roll back—is part of the method rather than a deployment afterthought.

The paper makes four scoped contributions. First, it formulates unseen-user ocular cleaning as corruption-only calibration with strictly disjoint support and query sets and an identifiable EOG-reference transfer operator. Second, it introduces a likelihood-ratio correction between population and context observation models that always condition on the same query. Third, it combines precision attenuation with a reliability-gated adaptive loop and an executable, pathwise rollback target. Fourth, it defines an intervention-based evaluation protocol that separates oracle mechanism, operator specificity, diffusion necessity, physiological preservation, and rollback. The instantaneous ridge/SVD projector is the only primary estimator. Robust, lagged, frequency-selective, electrode-graph, covariance-contrast, and population-shrunk alternatives are specified as a deferred portfolio to be activated only after a preregistered diagnostic failure, not as simultaneous routes for rescuing an unvalidated mechanism. We do not claim the first EEG diffusion model, the first calibration-based ocular correction method, the first subject-adaptive EEG model, or theoretical superiority of diffusion over the Bayes conditional mean.

The theoretical analysis is correspondingly limited but falsifiable. We prove exact cancellation of the modeled ocular term after applying the true complement, bound residual artifact and clean-signal distortion under projector error, and derive the exact score identity for a fixed positive observation potential before identifying the plug-in approximation used in practice. A truncated top- r result states the excitation and spectral-gap conditions under which the calibrated span is estimable. We also prove pathwise equality between the zero-gate implementation and an independently invoked population-posterior sampler. The geometric-density bridge applies only before state-dependent normalization, annealing, and clipping. No theorem orders an arbitrary wrong calibration against the population observation model, and no squared-error theorem can make a restoration draw dominate an exact conditional mean. Those questions are assigned to paired correct, population, wrong-user, shuffled, and oracle interventions; oracle-span orthogonal subtraction; native SGEYESUB; and two distinct one-step estimators.

The empirical design uses paired semi-simulated EEG/EOG for causal fidelity, real calibration recordings for attenuation and preservation, a multimodal multi-paradigm dataset for cross-session validation, and EEGdenoiseNet only as a generic stress test. Each montage receives its own population prior in this first implementation; the claim is replication of one method across datasets, not one checkpoint across incompatible channel spaces. Inference is performed at the participant or session level, not at the window level. Experiments proceed through an oracle mechanism gate, operator-specificity gate, diffusion-necessity gate, and only then learned attenuation, drift, and rollback. The deferred operator portfolio is opened only if the primary operator passes the mechanism test yet exhibits a registered failure mode. The protocol reports effect sizes and uncertainty together with overlap energy, latency, calibration burden, nominal interval coverage, drift detection, and rollback behavior. All numerical fields in this clean-room draft are explicit placeholders; no result is implied by the schematic figures or empty tables.

The remainder of the paper reviews classical, learned, and diffusion-based artifact removal and posterior sampling (Section 2); formalizes the held-out-context problem (Section 3); presents CSPD (Section 4); derives its guarantees and limitations (Section 5); specifies the falsification-oriented experiments (Section 6); and closes with limitations, responsible adaptation, and conclusions (Sections 7 and 8).

2 Related Work

2.1 Regression, source separation, and calibrated subspaces

Ocular correction methods differ in what they observe and in how they treat neural activity that overlaps the artifact. Early reference-regression methods estimate the propagation of electrooculography (EOG) into each EEG channel

and subtract the predicted contribution [8, 12, 34]. They are inexpensive and make the observation model explicit, but their coefficients can absorb frontal brain activity correlated with eye movements, while cap placement, impedance, gaze direction, and session state can change the transfer. Participant- or session-specific propagation coefficients and dedicated calibration movements are therefore established ideas, not contributions unique to this work.

Independent component analysis (ICA) instead assumes that ocular and neural activity can be separated into latent sources. It remains a standard offline baseline and can preserve activity outside the rejected components [18, 19]. Its central ambiguity is structural: an independent component may contain both ocular and neural activity, so rejecting it makes an irreversible decision about their overlap. Artifact Subspace Reconstruction (ASR) detects transient high-variance subspaces relative to a clean reference covariance and deterministically reconstructs affected samples with covariance-derived linear maps from retained directions. ASR is therefore a reconstruction method, not simple component zeroing, but it neither specifies a probabilistic clean-signal posterior nor represents reconstruction uncertainty. It supports online wearable EEG pipelines [30]; systematic evaluation and Riemannian covariance variants show that calibration and geometry materially affect the result [2, 4]. Multi-channel Wiener filtering provides a related supervised spatial-filter formulation [36]. These methods are indispensable baselines, but none couples current-session correction to a generative population prior over neural content that is uncertain within ocularly affected directions.

SGEYESUB is the closest calibration-based precursor [25]. It estimates sparse generalized ocular subspaces from an approximately five-minute calibration, after which the fitted correction reduces to a matrix multiplication. Its native correction is sequential: horizontal and vertical eye-movement components are corrected first, followed by residual blink and eyelid artifacts. Its 69-participant evaluation considers both artifact attenuation and preservation of event-related activity, and the reported within-session stability motivates a context-specific low-rank operator. Thus, neither short ocular calibration nor a participant-specific artifact subspace is new. This native sequential procedure must be distinguished from the oracle-span orthogonal-subtraction comparator used in our paired semi-simulation, which directly applies the exact orthogonal complement projector to the contaminated signal. In contrast, CSPD uses a support-estimated subspace to parameterize a context observation energy and invokes a separate population prior to reconstruct uncertain neural content. We consequently compare against both a faithful native SGEYESUB implementation where its calibration protocol is available and the matched orthogonal-subtraction control where the ground-truth span is known; population, wrong-participant, shuffled, and oracle operators then test the value of the estimated corruption geometry directly.

2.2 Deterministic learned EEG denoisers and physiological preservation

EEGdenoiseNet standardized paired, semi-synthetic evaluation by releasing clean EEG, ocular-artifact, and muscle-artifact segments [47]. It enabled direct waveform models ranging from residual convolutional networks [39] to embedding-space separation in DeepSeparator [46] and independent-component mixtures with IC-U-Net [5]. These models learn population mappings from corrupted to cleaned signals; their reconstruction scores do not by themselves establish adaptation to a previously unseen participant’s ocular operator.

CLEEGN targets subject-independent, plug-and-play reconstruction without new-user calibration. Its current citable record is the five-author arXiv v2 preprint, distinct from the shorter workshop author listing [26]. EEGOAR-Net is an archival 2025 journal article that trains a U-Net with SGEYESUB-derived targets and channel masking, yielding calibration-free operation across participants and montages without EOG at inference [29]. It is therefore the strongest direct counterpoint to personalization: calibration is useful only if a reliable held-out-user operator improves upon this population alternative without increasing neural distortion. Recent work also reduces supervision or computation.

iPSD and iPSD-Zero learn self-supervised partitions, including a single-segment zero-shot setting, but remain a 2026 preprint [33]. Task-Oriented Learning, by contrast, is now an archival 2026 *IEEE Transactions on Instrumentation and Measurement* article; it weights blind-source components using a downstream proxy-task loss rather than a clean reference [44]. BandRouteNet is a 2026 preprint that routes denoising across bands and temporal positions with a compact parameter budget [27], while EDGeNet explicitly targets efficient deployment [10].

A2DM belongs to this deterministic, artifact-aware family. Despite the “DM” suffix, it is *not* a diffusion model: it is an archival discriminative denoiser that fuses an artifact-type representation into frequency-selective removal and time-domain compensation [28]. It is relevant to mixed-artifact conditioning, not to diffusion posterior sampling. More broadly, a recent capacity-controlled preprint reports early saturation of EEGdenoiseNet reconstruction metrics and a gap between those metrics and downstream BCI utility [1]. This evidence motivates an equal-backbone, equal-capacity one-step predictor; parameter, latency, and sampler-step reporting; and fixed- as well as retrained-decoder evaluations. CSPD earns iterative inference only if it improves preregistered preservation or uncertainty criteria beyond that matched deterministic alternative.

2.3 Diffusion models for EEG

Denoising diffusion probabilistic models and score-based stochastic differential equations learn a sequence of noise-conditioned denoisers or scores that represents a data distribution rather than only a conditional mean [16, 38]. DDIM supplies a commonly used accelerated, partly deterministic sampling formulation [37]. EEGdfus adapts diffusion directly to paired EEG denoising: noisy EEG conditions a CNN–Transformer dual-branch network that generates the corresponding clean waveform [17]. D4PM, currently a 2025 preprint, trains complementary clean-EEG and artifact branches and combines their priors through joint posterior sampling with artifact-type information [35]. These studies establish diffusion-based EEG artifact removal and multi-branch posterior construction as prior art.

Other EEG diffusion work occupies adjacent personalization and restoration claims. DS-DDPM, for which no archival venue was verified, decomposes each denoising step into subject-domain variance and invariant content and supervises the domain component with a subject classifier [11]. Accordingly, “subject-specific EEG diffusion” is not an appropriate novelty claim for this paper. Conditional ERP synthesis already generates subject-, session-, and class-specific EEG with classifier-free guidance [24]. STAD and TopoDiff use conditional diffusion for spatial EEG super-resolution, respectively emphasizing spatio-temporal adaptation and electrode geometry or relations; both are preprints as of this review [40, 45]. They reconstruct missing channels rather than ocularly contaminated content, but demonstrate that conditional, subject-adaptive, and geometry-aware EEG diffusion are each established directions.

CSPD separates these ingredients differently. The clean-EEG score is trained at population level and never receives the test participant’s identity. A disjoint support recording estimates only a low-rank ocular corruption operator, which enters the query observation energy rather than the prior network. This separation permits controlled operator swaps and an auditable reliability gate. It also exposes hallucination risk: a restoration draw plausible under the population prior need not be the physiology that occurred. Measurement-constrained directions and sampler dispersion within attenuated directions must therefore be evaluated rather than treated as truth, and unreliable calibration must trigger population-posterior rollback rather than stronger guidance.

2.4 Diffusion posterior sampling and blind inverse problems

Diffusion restoration can be written as Bayesian inference with a learned data prior and a measurement likelihood. DDRM exploits the singular-value structure of a known linear operator [21]; DDNM enforces consistency in the operator

range while using the diffusion prior to refine null-space content [41]. The latter is particularly close mathematically because ocular subspace removal creates missing neural components that cannot be recovered by projection alone. Diffusion Posterior Sampling (DPS) handles general noisy linear and nonlinear observations by interleaving reverse diffusion with an approximate likelihood gradient [7], and DiffPIR embeds a generative denoiser in a plug-and-play restoration loop [48].

Unknown forward models are also prior art. BlindDPS runs parallel diffusion processes for the signal and operator [6], whereas GibbsDDRM samples an unknown linear operator and signal using a partially collapsed Gibbs construction [31]. Hence, neither posterior diffusion nor an estimated operator is sufficient novelty. Calibration makes the present problem deliberately less blind: it supplies an explicit low-dimensional operator estimate before query restoration, without learning an operator diffusion prior. The implemented sampler uses a DPS-like plug-in approximation. Its outputs are therefore plug-in guided restoration draws, not exact samples from a posterior that marginalizes operator-estimation or selector uncertainty. The EEG-specific contribution is the combination of disjoint calibration, a population-to-context observation-energy correction, reliability evidence, pathwise population-posterior rollback, and physiological validation.

2.5 Datasets, evaluation regimes, and the remaining gap

The relevant resources support different claims and should not be pooled as interchangeable benchmarks. The Klados-Bamidis resource provides paired pure and contaminated EEG for 27 participants, with participant- and electrode-specific VEOG/HEOG propagation coefficients [22, 23]. It is suited to a causal operator test under participant-held-out, disjoint support/query splits. Its four eye-region electrodes form two bipolar EOG signals; they are not four independent EOG channels. EEGdenoiseNet remains useful for generic EOG, EMG, mixed-artifact, and low-SNR stress tests, but its single-channel segment construction cannot substantiate an unseen-subject claim.

Real recordings require different endpoints. The Eye-BCI resource contains synchronized EEG, eye tracking, and high-speed video from 31 participants and 63 sessions across motor imagery, motor execution, SSVEP, and P300 paradigms [13, 14]. It supports cross-session and cross-paradigm artifact attenuation, neural preservation, and downstream utility, but has no perfect clean EEG target and therefore cannot support paired reconstruction error. EEGEyeNet adds broad simultaneous EEG/eye-tracking coverage, but is primarily an eye-movement-prediction benchmark rather than a clean/noisy denoising dataset [20].

Across these regimes, windows are repeated observations rather than independent participants. Comparisons should therefore aggregate to the participant level, retain train/test participant disjointness, and keep calibration support separate from every evaluated query. Paired data can support temporal and spectral error, correlation, and operator-recovery analyses. Natural data instead require convergent evidence: blink-locked residuals, EEG-EOG association, frontal topography, event-related amplitude and latency, oscillatory power, spatial covariance, and fixed- as well as retrained-decoder performance. Calibration duration and drift should be reported as experimental variables, not hidden preprocessing choices. These distinctions prevent a lower reconstruction error on synthetic mixtures from being interpreted as proof of physiological validity or practical personalization.

The remaining gap is consequently a conjunction rather than a claim over any single ingredient. Classical methods provide transparent regression, rejection, or deterministic covariance and subspace correction, but not a learned population distribution over uncertain clean EEG; population denoisers reconstruct signals but average over operators; existing EEG diffusion models provide rich priors or identity conditioning but do not confine unseen-user adaptation to a support-estimated ocular observation model; and generic inverse solvers do not define EEG calibration, rejection, or

Table 1. Conceptual positioning. “Context operator” denotes an explicit current-session corruption estimate. “Artifact-aligned treatment” distinguishes subtraction or rejection, deterministic correction or reconstruction, and generative restoration; these mechanisms should not be collapsed into a binary restoration label. Entries describe designs, not empirical rank.

Family	Population clean prior	Context operator	Artifact-aligned treatment	Reliability refusal	Operator-swap test
Regression / ICA	No	Partial / implicit	Subtraction / rejection	Rare	Limited
ASR	No	Calibration covariance	Deterministic covariance reconstruction	No explicit refusal	Limited
SGEYESUB calibration	No	Yes	Sequential subspace correction	No	Possible
Population neural denoiser	Implicit	No	Deterministic learned reconstruction	Rare	No
EEG diffusion cleaner	Yes	Usually implicit	Generative reconstruction	Rare	Limited
Generic diffusion inverse solver	Yes	Given or inferred	Prior-guided reconstruction	Not EEG-specific	Possible
CSPD	Yes	Yes, support-only	Plug-in prior-guided restoration	Pathwise zero-gate rollback	Required

preservation tests. CSPD targets this intersection while leaving its empirical value falsifiable through two one-step baselines and population, wrong/shuffled, oracle, and subtraction controls.

3 Problem Formulation

3.1 Contexts, support, and unseen-user queries

Let a deployment context $\omega = (s, r, g)$ identify a user s , recording session r , and acquisition state g comprising the montage, reference convention, and electrode condition. The context is deliberately broader than immutable identity: replacing a cap or changing the reference can alter ocular propagation for the same person. A population collection $\mathcal{D}_{\text{train}}$ contains contexts from users that are disjoint from every test user. It is used to learn a clean-EEG prior, a population observation operator, an EEG-side reliability estimator, and a mapping from calibration diagnostics to a context reliability. None of these population parameters is updated on a test query.

For an unseen context ω^* , the system first receives a short calibration support

$$C_{\omega^*} = \{(Y_i, E_i, \ell_i)\}_{i=1}^{n_c}, \quad (1)$$

where $Y_i \in \mathbb{R}^{C \times T_i}$ is EEG, $E_i \in \mathbb{R}^{d \times T_i}$ contains synchronized EOG channels or prespecified ocular-reference regressors, and ℓ_i records prompted movements and acquisition metadata. The query collection $Q_{\omega^*} = \{Y_q\}_{q=1}^{n_q}$ contains the subsequent EEG windows. Its times, trials, and events are disjoint from the support. Query EOG, eye tracking, or a manual ocular annotation can supply an external attenuation trace in a reference experiment. The primary EEG-only

setting instead obtains that trace from a detector trained on population contexts; the two settings have different probabilistic interpretations in Section 4.

This protocol is neither zero-shot inference nor transductive fitting on the test query. It is supervised sensor calibration from a dedicated support: the system estimates a low-dimensional observation operator, evaluates support-only diagnostics, and freezes the accepted operator until monitoring requests rollback or a new calibration. Performance is therefore reported as a function of support duration and includes the acquisition cost.

3.2 Identifiable EOG-reference transfer

For a C -channel, T -sample window in context ω , the modeled observation is

$$Y = X + C_\omega E + \Xi, \quad Y, X, \Xi \in \mathbb{R}^{C \times T}, \quad C_\omega \in \mathbb{R}^{C \times d}, \quad E \in \mathbb{R}^{d \times T}. \quad (2)$$

Here X is clean neural EEG, E is expressed in the same specified EOG-reference coordinates used during calibration, and C_ω transfers those coordinates to the EEG sensors. Thus the fitted matrix is an effective reference-to-EEG transfer, rather than a factorization into an unobserved ocular source and a non-identifiable mixing matrix. Components not expressed by the retained reference coordinates, non-ocular artifacts, sensor noise, and within-context drift are included in Ξ .

Only reference directions excited by the support are identifiable. Let $\mathcal{R}_E \subseteq \mathbb{R}^d$ be the regressor subspace retained by a training-frozen rule. Calibration and all claims about C_ω are restricted to $\mathcal{R}_E$; unexcited directions are not recovered by regularization. After this restriction, the modeled transfer has effective rank $r \ll C$. Its spatial projector and complement are

$$\Pi_\omega = C_\omega (C_\omega^\top C_\omega)^\dagger C_\omega^\top = U_\omega U_\omega^\top, \quad Q_\omega = I_C - \Pi_\omega, \quad (3)$$

where $U_\omega \in \mathbb{R}^{C \times r}$ is an orthonormal basis for the retained transfer range. Consequently, $\Pi_\omega = \Pi_\omega^\top = \Pi_\omega^2$. The implementation obtains U_ω from a rank- r truncated SVD and applies $U_\omega (U_\omega^\top v)$ without forming a dense projector.

For comparison, the regularized matrix

$$S_\varepsilon(C) = C(C^\top C + \varepsilon I)^{-1} C^\top, \quad \varepsilon > 0, \quad (4)$$

is a positive-semidefinite shrinkage operator, not an orthogonal projector. It is generally non-idempotent and does not cancel CE through $I - S_\varepsilon$. Exact projector statements below therefore use $\widehat{\Pi} = \widehat{U}_r \widehat{U}_r^\top$; S_ε is confined to a leakage sensitivity analysis.

When a spatial operator acts at every time point, its column-vectorized form is $I_T \otimes \Pi_\omega$. We retain matrix notation to expose the spatial geometry. Norms without qualification are Euclidean for vectors, Frobenius for signal matrices, and spectral for linear operators.

3.3 Population and context observation energies

The same corrupted query $y = \text{vec}(Y)$ conditions both sides of the adaptation gate. Training contexts determine a population observation energy $\mathcal{E}_0(x; y)$ with precision $\mathcal{W}_0$, while the accepted support C_ω determines a context energy $\mathcal{E}_C(x; y)$ with precision $\mathcal{W}_C$. Reliability $\rho_\omega \in [0, 1]$ interpolates the energies,

$$\mathcal{E}_\rho(x; y) = (1 - \rho_\omega) \mathcal{E}_0(x; y) + \rho_\omega \mathcal{E}_C(x; y), \quad \mathcal{W}_\rho = (1 - \rho_\omega) \mathcal{W}_0 + \rho_\omega \mathcal{W}_C. \quad (5)$$

The clean-EEG prior is population trained in both cases. Hence $\rho_\omega = 0$ does not discard y and does not invoke an unconditional generator: it selects the population posterior, or the corresponding population generalized-Bayes

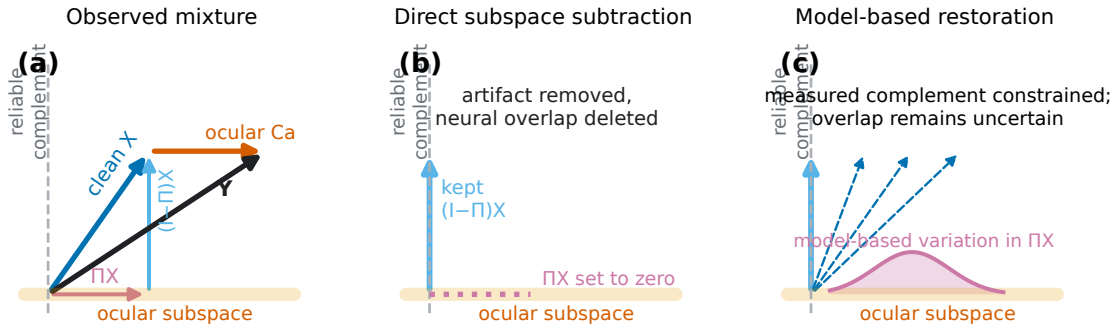


Fig. 1. Subspace geometry of ocular cleaning. Direct projection removes the reference-aligned contamination but also deletes neural activity in the same spatial span. The proposed observation energy weights the measured complement more strongly and uses a population clean-EEG prior to form model-based reconstructions in the attenuated span. The soft constraint does not assert samplewise equality with the observation.

update in the EEG-only-mask setting, conditioned on that same y . NULL is implemented as an explicit call to this population-posterior path.

3.4 Why subtraction is insufficient

If the retained transfer is exact and $\Xi = 0$, direct subspace subtraction returns

$$Q_{\omega} Y = Q_{\omega} X. \quad (6)$$

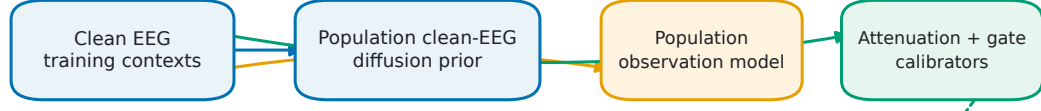
The modeled reference-aligned artifact vanishes, but so does the legitimate component $\Pi_{\omega} X$. Reconstructing $Y - \widehat{CE}$ has the same ambiguity when an estimated artifact amplitude absorbs neural activity. The reliable complement is identifiable under the model; the missing neural coordinate is not identifiable from the corrupted window alone and requires a prior.

Figure 1 contrasts subtraction with model-based restoration. Subtraction assigns zero to the removed coordinate. CSPD instead gives the measured complement greater observation weight and lets the population prior represent ambiguity in the attenuated coordinate. This is a soft finite-precision constraint: it penalizes, but does not algebraically guarantee, agreement with the query complement. Its stochastic reconstructions and empirical dispersion are therefore evaluated as model outputs, not interpreted as recovered physiological ground truth or certified draws from an exact conditional law.

3.5 Design requirements

The formulation imposes five requirements. Calibration and query must be disjoint so that leakage cannot masquerade as personalization. The context operator must alter only the observation energy, not the population prior. The soft energy must expose how strongly each coordinate is weighted rather than claim exact preservation. Reliability must be learned and audited on held-out training contexts. Finally, invalid calibration must short-circuit to the population observation energy for the same query, so NULL reproduces the population-posterior algorithm rather than a zero-valued context branch.

Population stage (training users only)



Unseen deployment context

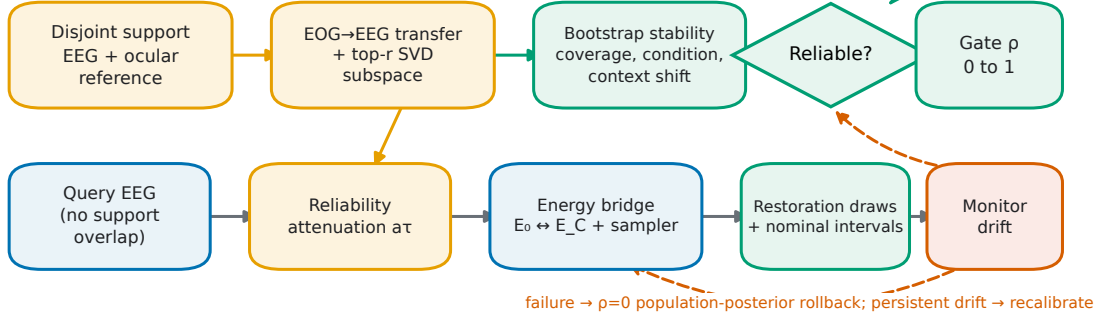


Fig. 2. CSPD adaptive-system loop. Training contexts determine the clean-EEG prior and population observation energy. A query-disjoint EEG–EOG support determines a context energy. Reliability interpolates the two energies for the same query; rejection short-circuits to the population-posterior procedure.

4 Calibration-Guided Subspace Posterior Diffusion

Figure 2 shows the adaptive loop. Population training learns a clean-EEG prior, a montage-compatible population observation operator, an EEG-side attenuation estimator, and reliability calibration. A disjoint support estimates only the current EOG-reference transfer and its diagnostics. Query inference combines the population and context observation energies for the same corrupted window. Invalid calibration never removes observation conditioning; it invokes the population-posterior path.

4.1 Population clean-EEG diffusion prior

Let $x_0 = \text{vec}(X) \in \mathbb{R}^{CT}$. A variance-preserving forward process gives

$$q(x_t | x_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I\right), \quad t \in \{1, \dots, T_d\}, \quad (7)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{j=1}^t \alpha_j$. A multichannel temporal network is trained with

$$\mathcal{L}_{\text{prior}}(\theta) = \mathbb{E}_{x_0, t, \epsilon} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c_g) \right\|_2^2 \right], \quad (8)$$

where $\epsilon \sim \mathcal{N}(0, I)$. The optional acquisition condition c_g is a dataset-specific, fixed-montage key together with the registered reference convention and sampling rate; it is not a learned canonical map and contains no person identity, test support, or query. Training contexts are grouped by user and exclude all evaluation users.

The population score and Tweedie estimate are

$$s_\theta(x_t, t, c_g) = -\epsilon_\theta(x_t, t, c_g) / \sqrt{1 - \bar{\alpha}_t}, \quad (9)$$

$$\hat{x}_0(x_t, t) = \frac{x_t + (1 - \bar{\alpha}_t)s_\theta(x_t, t, c_g)}{\sqrt{\bar{\alpha}_t}}. \quad (10)$$

This network represents an unconditional clean-EEG prior. Observation conditioning is introduced only through the population or context energy below. Consequently, the population fallback remains conditioned on y through $\mathcal{E}_0$ rather than reverting to an unconditional generation trajectory.

4.2 Reference-space restriction and top-rank calibration

After alignment and preprocessing frozen from training contexts, concatenate the centered support into $Y_c \in \mathbb{R}^{C \times N}$ and $E_c \in \mathbb{R}^{d \times N}$. Let $V_R \in \mathbb{R}^{d \times q}$ have orthonormal columns spanning the regressor space retained by a training-frozen excitation rule, and define

$$\tilde{E}_c = V_R^\top E_c, \quad G_R = \tilde{E}_c \tilde{E}_c^\top \succ 0, \quad C_{\omega, R} = C_\omega V_R. \quad (11)$$

Positive definiteness is required only on this retained q -dimensional space. Directions removed by the excitation rule are declared unidentified and folded into the mismatch term; ridge regularization is not used to manufacture information in them.

The primary calibration estimator is

$$\hat{C}_{\lambda, R} = \arg \min_{C \in \mathbb{R}^{C \times q}} \left\| Y_c - C \tilde{E}_c \right\|_F^2 + \lambda \|C\|_F^2 = Y_c \tilde{E}_c^\top (G_R + \lambda I_q)^{-1}. \quad (12)$$

The implementation solves the corresponding linear system and does not form the displayed inverse. Let

$$\hat{C}_{\lambda, R} = \hat{U} \hat{\Sigma} \hat{V}^\top, \quad \hat{\Pi}_C = \hat{U}_{:,1:r} \hat{U}_{:,1:r}^\top, \quad \hat{Q}_C = I_C - \hat{\Pi}_C. \quad (13)$$

This is explicitly a top- r truncated SVD. The retained regressor rule, λ , r , and any singular-gap threshold are frozen on training contexts; query outcomes cannot change them.

Training contexts also produce a montage-, reference-, and preprocessing-specific population state Ω_0 , including a population projector $\hat{\Pi}_0$ or, for a registered non-projector family, a population precision operator. It is estimated without the held-out user's support. The context state Ω_C contains $\hat{\Pi}_C$ and its support-only diagnostics. Block bootstrap projectors $\hat{\Pi}_C^{(b)}$ yield

$$u_{\text{stab}} = \text{median}_{b < b'} \left\| \hat{\Pi}_C^{(b)} - \hat{\Pi}_C^{(b')} \right\|_2. \quad (14)$$

Conditioning of G_R , retained excitation, prompted-movement coverage, effective block count, held-out support residuals, and a target-blind support-query distance complete the diagnostic record. Query summaries can reduce reliability but cannot refit the operator.

4.3 Precision attenuation inside the ocular span

Let $a_\tau \in [0, 1]$ be a precision-factor reliability at query sample or frame τ . It is not interpreted as an artifact probability. The training-frozen mapping is monotone in evidence of ocular contamination: $a_\tau = 1$ assigns the same observation amplitude to all spatial directions, while $a_\tau = 0$ removes observation precision from the selected ocular span. For projector state $j \in \{0, C\}$, define

$$M_{j, \tau} = Q_j + a_\tau \Pi_j, \quad W_{j, \tau} = M_{j, \tau}^\top M_{j, \tau} = Q_j + a_\tau^2 \Pi_j. \quad (15)$$

The square in $W_{j,\tau}$ is essential: a_τ attenuates a precision factor, not precision itself.

With validity weights w_τ , the population and context energies are

$$\mathcal{E}_j(x_0; y) = \frac{1}{2\sigma_y^2} \sum_{\tau=1}^T w_\tau (x_{0,\tau} - y_\tau)^\top W_{j,\tau} (x_{0,\tau} - y_\tau), \quad j \in \{0, C\}. \quad (16)$$

Reliability ρ_ω then gives Equation (5), equivalently

$$W_{\rho,\tau} = (1 - \rho_\omega)W_{0,\tau} + \rho_\omega W_{C,\tau}. \quad (17)$$

Finite σ_y^2 makes this a soft observation penalty. Even on $\text{range}(Q_j)$, a reconstruction is not asserted to equal y exactly; complement agreement and neural preservation remain measured outcomes.

Registered but deferred backup families use the same interface with positive-semidefinite operators $W_{j,\tau}$. Projector symmetry, idempotence, cancellation, and principal-angle bounds apply only when a family returns an orthogonal spatial projector. A covariance or filter-bank precision is not renamed a projector to inherit those results.

4.4 External-mask likelihood versus EEG-only generalized Bayes

When a_τ is supplied by an external modality or annotation and treated as an observed covariate, Equation (16) has a conditional Gaussian interpretation. If the externally indexed nuisance variance in the ocular span is κ_τ^2 , the precision-factor amplitude is

$$a_\tau = \left(\frac{\sigma_y^2}{\sigma_y^2 + \kappa_\tau^2} \right)^{1/2}. \quad (18)$$

For $a_\tau > 0$,

$$y_\tau \mid x_\tau, a_\tau, \Pi_j \sim \mathcal{N}(x_\tau, \sigma_y^2 Q_j + \sigma_y^2 a_\tau^{-2} \Pi_j), \quad (19)$$

whose precision is $\sigma_y^{-2} W_{j,\tau}$. At $a_\tau = 0$, the ordinary Gaussian density is defined only for the observed coordinate $U_{j,\perp}^\top y_\tau$; the ocular coordinate is omitted. The Gaussian normalizer is constant in x for a supplied a_τ and operator, but it must be retained when fitting attenuation or variance parameters by likelihood.

In the primary EEG-only mode, $a_\tau = h_\phi(Y)_\tau$ is computed from the same EEG window that appears as y . Holding this value constant while updating x gives a valid differentiable loss, but $\exp[-\mathcal{E}_j(x; y)]$ is not claimed to be a normalized likelihood over y : the detector's dependence on y changes that normalizer and selection mechanism. We therefore define the generalized-Bayes target

$$\pi_\rho^{\text{GB}}(x_0 \mid y) \propto p_X(x_0) \exp[-\mathcal{E}_\rho(x_0; y)] \quad (20)$$

and call its outputs guided stochastic reconstructions. In the external-mask mode the analogous expression is an ordinary posterior under Equation (19); in the EEG-only mode it is a Gibbs update induced by a frozen loss. The algorithm and gradient are shared, but the semantic claim is not.

4.5 Fixed-energy identity and plug-in guidance

Let $L_\rho(y; x_0) = \exp[-\mathcal{E}_\rho(x_0; y)]$. It is a likelihood factor in the external-mask mode and a compatibility factor in the EEG-only mode. In either case, the normalized fixed-energy target at noise level t obeys the exact algebraic identity

$$\pi_{\rho,t}(x_t \mid y) \propto p_t(x_t) \mathbb{E}[L_\rho(y; X_0) \mid X_t = x_t], \quad (21)$$

and hence

$$\nabla_{x_t} \log \pi_{\rho,t}(x_t | y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log \mathbb{E}[L_\rho(y; X_0) | X_t = x_t]. \quad (22)$$

Calling this a Bayes posterior score is reserved for the external-mask likelihood. For the EEG-only detector it is the score of the specified generalized posterior.

The conditional expectation is generally intractable. The implemented raw correction evaluates the energy at the Tweedie estimate:

$$g_{\rho,t}^{\text{raw}} = -\gamma_t J_{\widehat{x}_0}(x_t, t)^\top \nabla_{x_0} \mathcal{E}_\rho(\widehat{x}_0(x_t, t); y), \quad (23)$$

$$\nabla_{x_{0,\tau}} \mathcal{E}_\rho = \frac{w_\tau}{\sigma_y^2} W_{\rho,\tau}(x_{0,\tau} - y_\tau). \quad (24)$$

Automatic differentiation computes the vector-Jacobian product. Replacing the Jacobian by an identity map or detaching the denoiser is a separate ablation. The implemented score passed to the reverse update is

$$s_\rho^{\text{impl}}(x_t, t) = s_\theta(x_t, t, c_g) + \mathcal{T}_t(g_{\rho,t}^{\text{raw}}), \quad (25)$$

where γ_t supplies the frozen annealing schedule and $\mathcal{T}_t$ applies residual-dimension normalization, norm normalization, and trust-radius clipping. Equations (23)–(25) define an approximate guided reverse process, not an exact posterior sampler.

Before $\mathcal{T}_t$ normalizes or clips the correction, energy linearity gives

$$g_{\rho,t}^{\text{raw}} = (1 - \rho_\omega) g_{0,t}^{\text{raw}} + \rho_\omega g_{C,t}^{\text{raw}}. \quad (26)$$

At the clean level, the corresponding unnormalized factors satisfy $p_X e^{-\mathcal{E}_\rho} = (p_X e^{-\mathcal{E}_0})^{1-\rho} (p_X e^{-\mathcal{E}_C})^\rho$. An analogous unnormalized plug-in identity holds at a fixed noise level before guidance-vector normalization and clipping. We make no density-bridge claim after $\mathcal{T}_t$, and forward noising does not generally commute with this geometric interpolation.

4.6 Reliability, monitoring, and population-posterior rollback

For the primary operator, let

$$z_\omega = (u_{\text{stab}}, \kappa_R, c_{\text{cover}}, n_{\text{eff}}, d_{sq}, q_a) \quad (27)$$

collect subspace instability, retained-regressor conditioning, prompted-movement coverage, effective support size, support-query distance, and attenuation-estimator quality. A monotone calibrator fitted only on training contexts returns

$$\rho_\omega = g_\psi(z_\omega) \in [0, 1]. \quad (28)$$

Rank failure, missing required movements, sensor-map mismatch, nonfinite output, or distance outside training support sets $\rho_\omega = 0$. The initial confirmatory protocol is P0-or-NULL. B1–B6 are not searched automatically: after the RQ1 and RQ2 mechanism gates have passed, at most one family may be activated by a prospective amendment that freezes its triggering diagnostic and moves evaluation to a new untouched outer split. A valid P0 estimate is never displaced by a backup, and query targets never participate.

NULL and every explicit $\rho_\omega = 0$ decision short-circuit before a context energy is built and call the population-posterior procedure with $\mathcal{E}_0$, the same y , and the registered random stream. This explicit branch is required because nonlinear normalization, clipping, random-number consumption, or a malformed zero operator could otherwise make a nominal zero gate differ from the population algorithm.

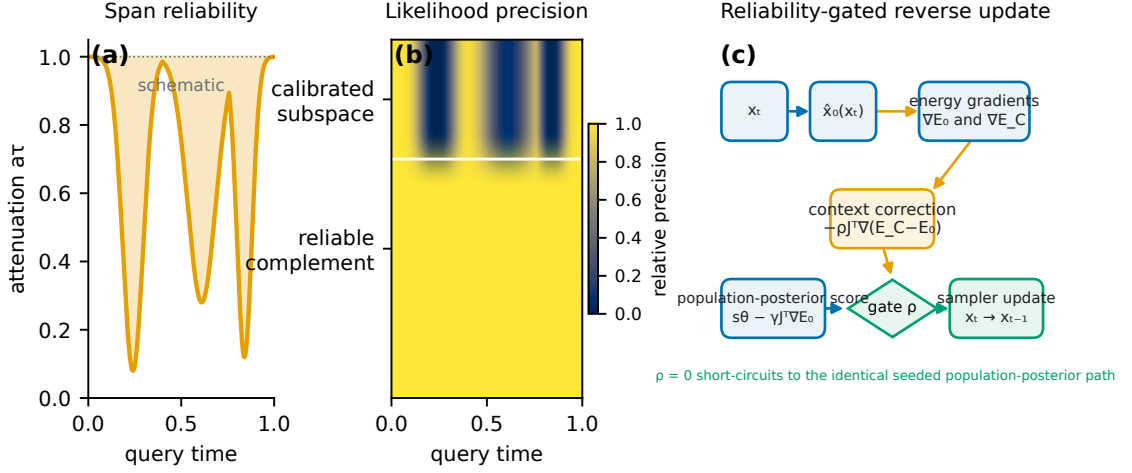


Fig. 3. Energy-guided restoration. Reliability-amplitude decay controls precision in an ocular span, while context reliability interpolates a population energy and a support-calibrated energy for the same query. A vector-Jacobian product transports the raw energy gradient; normalization and clipping are implementation operations and are not assigned a density-bridge interpretation.

Monitoring can lower ρ_ω or request a new support episode but cannot change the selected operator family within a reverse trajectory. A rejection after context-guided updates restarts from the saved initial state and random stream through the population-posterior branch. Reported uncertainty outputs are restoration draws, empirical dispersion, and nominal quantile bands; posterior or coverage semantics require a separate held-out calibration result.

4.7 Computation and implementation boundary

Calibration forms $Y_c \tilde{E}_c^T$ in $O(CqN)$ time and decomposes a $C \times q$ matrix in $O(Cq^2)$ time when $q \ll C$. Storing a basis costs $O(Cr)$, and applying $Uv = U(U^T v)$ costs $O(Cr)$ per frame. Query cost is dominated by prior evaluations and vector-Jacobian products. We report calibration time, function evaluations, latency by window and channel count, peak memory, and performance as the number of reverse updates is reduced. A parameter-, receptive-field-, and information-matched one-step estimator is the primary computational control.

Under the primary family, test-time state contains the population state Ω_0 , accepted context basis $\hat{U}_r$, diagnostics, ρ_ω , attenuation-estimator version, monitor state, and acquisition metadata. Network weights remain population parameters. Raw support EEG and EOG can be deleted after approved operator and audit summaries are produced. Population and context energy versions, the NULL short-circuit, normalization and clipping rules, and random-stream identifiers are logged so that rollback can be reproduced.

5 Analysis

This section separates consequences of the stated transfer model from approximations introduced by truncation, EEG-derived attenuation, plug-in guidance, and clipping. Complete derivations appear in Appendix A.

5.1 Finite calibration on the identifiable regressor space

Let $\tilde{E}_c = V_R^\top E_c$ be the retained reference matrix from Equation (11), and write

$$Y_c = C_{\omega,R} \tilde{E}_c + N_c, \quad G_R = \tilde{E}_c \tilde{E}_c^\top \succ 0, \quad D_\lambda = G_R + \lambda I_q. \quad (29)$$

The positive-definiteness assumption concerns only the retained regressor space. The method makes no identifiability claim for $\ker(G)$ in the original EOG coordinates. Substitution into Equation (12) gives

$$\hat{C}_{\lambda,R} - C_{\omega,R} = N_c \tilde{E}_c^\top D_\lambda^{-1} - \lambda C_{\omega,R} D_\lambda^{-1}. \quad (30)$$

The first term contains neural-EOG correlation, sensor noise, and model mismatch; calling the support supervised does not make it mean zero. The second term is numerical ridge bias in transfer coordinates.

For the subspace result, define

$$R_\lambda = G_R (G_R + \lambda I_q)^{-1}, \quad A_\lambda = C_{\omega,R} R_\lambda, \quad Z_\lambda = N_c \tilde{E}_c^\top D_\lambda^{-1}, \quad \hat{C}_{\lambda,R} = A_\lambda + Z_\lambda. \quad (31)$$

Let $\Pi_{\lambda,r}$ and $\hat{\Pi}_r$ project onto the leading r left singular subspaces of A_λ and $\hat{C}_{\lambda,R}$, respectively, and let

$$\Delta_{\lambda,r} = \sigma_r(A_\lambda) - \sigma_{r+1}(A_\lambda) > 0, \quad e_\lambda = \|Z_\lambda\|_2. \quad (32)$$

With $\sigma_{q+1} = 0$ when needed, Wedin's $\sin-\Theta$ theorem [42] and Weyl's inequality give, whenever $e_\lambda < \Delta_{\lambda,r}$,

$$\|\hat{\Pi}_r - \Pi_{\lambda,r}\|_2 \leq \min \left\{ 1, \frac{\sqrt{2} e_\lambda}{\Delta_{\lambda,r} - e_\lambda} \right\}. \quad (33)$$

This is a top- r result, not the earlier full-column-rank shortcut. If $C_{\omega,R}$ has rank exactly r , R_λ is invertible because $G_R \succ 0$, so A_λ and $C_{\omega,R}$ have the same complete column range and $\Pi_{\lambda,r} = \Pi(C_{\omega,R})$. If the transfer has a nonzero tail beyond r , Equation (33) targets the regularized leading subspace. Comparison with the unregularized top- r target must additionally report

$$\|\hat{\Pi}_r - \Pi_r(C_{\omega,R})\|_2 \leq \underbrace{\|\hat{\Pi}_r - \Pi_{\lambda,r}\|_2}_{\text{ridge-truncation target shift}} + \underbrace{\|\Pi_{\lambda,r} - \Pi_r(C_{\omega,R})\|_2}_{\text{ridge-truncation target shift}}. \quad (34)$$

Thus sufficient excitation, a retained singular gap, and a perturbation smaller than that gap are substantive conditions. Bootstrap stability is an empirical proxy, not a proof that they hold.

The shrinkage matrix in Equation (4) has eigenvalues $\sigma_i^2 / (\sigma_i^2 + \varepsilon)$ on the retained span. Hence

$$\|\Pi(C) - S_\varepsilon(C)\|_2 = \frac{\varepsilon}{\sigma_r(C)^2 + \varepsilon}, \quad \|S_\varepsilon^2 - S_\varepsilon\|_2 = \max_i \frac{\varepsilon \sigma_i^2}{(\sigma_i^2 + \varepsilon)^2}. \quad (35)$$

Numerical shrinkage therefore cannot support exact cancellation claims.

5.2 Reliable complement, subtraction, and projector error

For this subsection, C_ω denotes the modeled retained transfer after any omitted tail has been placed in Ξ , $\Pi = \Pi(C_\omega)$, and $Q = I - \Pi$.

THEOREM 5.1 (RELIABLE-COMPLEMENT IDENTITY). *Under Equation (2),*

$$QY = QX + Q\Xi. \quad (36)$$

The complement cancels the modeled reference-aligned term, but it is neither noise-free nor the complete clean EEG.

Direct subtraction $X_{\text{sub}} = QY$ obeys

$$X_{\text{sub}} - X = -\Pi X + Q\Xi, \quad \|X_{\text{sub}} - X\|_F^2 = \|\Pi X\|_F^2 + \|Q\Xi\|_F^2. \quad (37)$$

The equality follows from spatial orthogonality and identifies the neural component deleted by subtraction. It does not assert that a learned prior can recover the realized value of that component.

Let $\widehat{\Pi}$ be an estimated rank-matched projector and

$$\delta_{\Pi} = \|\widehat{\Pi} - \Pi\|_2 = \sin \theta_{\max} \in [0, 1]. \quad (38)$$

Then

$$\|(I - \widehat{\Pi})C_{\omega}E\|_F \leq \delta_{\Pi} \|C_{\omega}E\|_F, \quad (39)$$

$$\|(I - \widehat{\Pi})X - QX\|_F \leq \delta_{\Pi} \|X\|_F. \quad (40)$$

The same projector error controls residual modeled artifact and perturbation of the nominal complement. Replacing $\widehat{\Pi}$ by $S_{\varepsilon}(\widehat{C}_r)$, where $\widehat{C}_r$ is the same rank- r truncation used to form $\widehat{\Pi}$, adds $\varepsilon/(\widehat{\sigma}_r^2 + \varepsilon)$ to the operator bound.

5.3 What the attenuation energy means

For a projector state, Equation (15) gives

$$W_{j,\tau} = Q_j + a_{\tau}^2 \Pi_j. \quad (41)$$

When a_{τ} is an externally supplied covariate and $a_{\tau} > 0$, this is the dimensionless precision of the proper Gaussian likelihood in Equation (19). At $a_{\tau} = 0$, $U_{j,\perp}^{\top} y_{\tau}$ has an ordinary $(C - r)$ -dimensional Gaussian density and the ocular coordinate is not observed. A proper clean-EEG prior then yields a proper posterior under standard integrability conditions.

When $a_{\tau} = h_{\phi}(Y)_{\tau}$ is computed from the same EEG, treating its realized value as constant makes the gradient in Equation (24) well defined, but does not turn the energy into a normalized likelihood over Y . The correct term is generalized-Bayes energy unless a joint model for $(Y, h_{\phi}(Y))$ is supplied. This distinction does not change the matrix derivative; it changes the inferential claim.

Moreover, finite energy is not equality constrained inference. Even Q_j receives finite precision σ_y^{-2} , so neither the posterior nor the generalized posterior is supported only on $Q_j x = Q_j y$. Statements that the soft method “exactly preserves” or “keeps fixed” the complement would therefore be false. Agreement is encouraged and must be quantified experimentally.

PROPOSITION 5.2 (FIXED-ENERGY OPERATOR INTERFACE). *Let $W_{\omega,j} \succeq 0$ be any finite operator held constant with respect to the latent state during an inference run, and set $\mathcal{E}_{\omega,j}(x; y) = \frac{1}{2}(x - y)^{\top} W_{\omega,j}(x - y)$. Then*

$$\nabla_x \mathcal{E}_{\omega,j}(x; y) = W_{\omega,j}(x - y). \quad (42)$$

Positive-definite W can parameterize an anisotropic Gaussian likelihood. Singular W can parameterize a likelihood on its observed range. If W is constructed from the same y without a joint observation model, the same expression is instead a generalized-Bayes loss. None of these cases implies that W is a projector.

5.4 Exact fixed-energy identity and approximate computation

Equation (22) is exact for the normalized target induced by the positive factor $L_\rho = e^{-\mathcal{E}_\rho}$. The implemented correction is not exact because

$$\nabla_{x_t} \log \mathbb{E}[L_\rho(y; X_0) \mid x_t] \neq \nabla_{x_t} \log L_\rho(y; \mathbb{E}[X_0 \mid x_t]) \quad (43)$$

in general. The vector-Jacobian product in Equation (23) is the full chain rule for this plug-in approximation. Denoiser detachment, $\gamma_t \neq 1$, guidance normalization, and clipping are further approximations.

Before normalization and clipping, Equation (26) follows exactly from the linear interpolation of energies. At a fixed noise level the raw plug-in vector field is the gradient of the unnormalized surrogate

$$\tilde{q}_{\rho,t}(x_t) = p_t(x_t) \exp[-\gamma_t \mathcal{E}_\rho(\hat{x}_0(x_t, t); y)]. \quad (44)$$

Its unnormalized factor is the geometric interpolation of the corresponding population and context factors. Once the raw correction is norm-normalized or clipped, it need not be conservative, so no density-bridge interpretation is asserted. Nor are the exact forward-noised population and context targets generally related by geometric interpolation.

5.5 Population-posterior fallback and bounded influence

THEOREM 5.3 (PATHWISE POPULATION-POSTERIOR FALLBACK). *Suppose NULL and $\rho_\omega = 0$ invoke the registered population-posterior procedure before context-operator construction. If the rollback and direct population runs use the same query y , population energy $\mathcal{E}_0$, initial reverse state, update implementation, and random stream, their trajectories and outputs are identical.*

This result is architectural rather than statistical. It does not say that $\mathcal{E}_0$ is the true observation model or that the reliability estimator rejects every harmful context. Explicit short-circuiting matters because a nominal zero multiplication after a context-specific normalization, clipping, or random operation need not reproduce the population code path.

A local bounded-influence statement remains possible. Let the continuous interpolation of the population procedure be $\dot{x}_0 = f_0(x_0, t)$, with f_0 $L(t)$ -Lipschitz. Suppose the normalization-and-clipping map is $L_{\mathcal{T}}(t)$ -Lipschitz on the visited region and $\|g_{C,t}^{\text{raw}} - g_{0,t}^{\text{raw}}\|_2 \leq G(t)$. Energy interpolation then gives a perturbation no larger than $\rho L_{\mathcal{T}}(t)G(t)$, and Gronwall's inequality yields

$$\|x_\rho(t) - x_0(t)\|_2 \leq \rho \int_{t_0}^t L_{\mathcal{T}}(u)G(u) \exp\left(\int_u^t L(v) dv\right) du. \quad (45)$$

This controls trajectory sensitivity under stated local assumptions; it is not a restoration-risk bound.

5.6 Bayes risk and the role of stochastic reconstruction

Let K be the permitted calibration summary and let both the calibrated and null decisions observe the same Y . Under the true joint distribution, the optimal proper-log-score risks satisfy

$$\mathcal{R}_{\log}^0 - \mathcal{R}_{\log}^K = H(X \mid Y) - H(X \mid Y, K) = I(X; K \mid Y) \geq 0. \quad (46)$$

For squared loss, with $\mu_0 = \mathbb{E}[X \mid Y]$ and $\mu_K = \mathbb{E}[X \mid Y, K]$,

$$\mathcal{R}_2^0 - \mathcal{R}_2^K = \mathbb{E} \|\mu_K - \mu_0\|_F^2 \geq 0. \quad (47)$$

These are non-strict population results for correct conditioning. They do not certify the learned energies, and finite estimation can reverse empirical ordering. If a shuffled summary K' is conditionally independent of X given Y , an ideal

method ignores it and equals null. No distribution-free theorem orders a plausible wrong operator and the population operator.

Finally, $\mu(Z) = \mathbb{E}[X | Z]$ minimizes squared error among estimators with inputs Z . One independent draw from the exact posterior has conditional MSE $2 \text{tr Cov}(X | Z)$, whereas the conditional mean has $\text{tr Cov}(X | Z)$. This ideal calculation does not describe the implemented clipped reverse process, whose outputs are called stochastic reconstructions. Diffusion must earn its role through finite-model performance, useful empirical dispersion, non-quadratic downstream decisions, or a better attenuation–preservation frontier. Quantile bands from repeated reconstructions are reported as empirical bands unless coverage calibration justifies stronger terminology.

6 Experimental Design

This section specifies a falsification-oriented protocol rather than reporting completed experiments. The protocol is frozen before any test-context result is inspected, and every numerical result remains a [TBD: provenance-tracked empirical value]. The central unit of generalization is a deployment context—a participant, session, montage, and electrode state—rather than a randomly sampled window. Paired targets and oracle operators are used only where the data-generating construction makes them available. Natural recordings are evaluated through artifact attenuation, physiological preservation, downstream utility, predictive calibration, and rollback criteria; they are not assigned an unobservable “clean” waveform.

6.1 Research questions and preregistered gates

RQ1: Does the oracle mechanism recover neural activity that orthogonal subtraction deletes? The first experiment uses paired semi-simulation and freezes every avoidable nuisance: the data-generating ocular span, a hard oracle event indicator, the observation variance, and the query windows. It compares the population posterior, the oracle-calibrated posterior, oracle-span orthogonal subtraction, and an information-matched one-step estimator. The primary outcome is error in the oracle ocular span, $\|\Pi(\hat{X} - X)\|_F$, while $\|Q(\hat{X} - Y)\|_F$ and clean-interval change are simultaneous measurement-consistency checks. We also report the clean-signal overlap ratio $r_{\text{overlap}} = \|\Pi X\|_F^2 / \|X\|_F^2$: a negligible ratio would limit the practical value of restoring a mathematically valid overlap. The gate passes only if oracle-calibrated restoration improves the preregistered span-specific outcome over oracle-span subtraction and the population posterior by [TBD: effect and confidence rule] without violating [TBD: complement and clean-interval margins]. Failure stops the personalized-diffusion program before learned detectors or backup operators are introduced.

RQ2: Is any benefit specific to the current context’s transfer operator? All conditions retain the same prior, query, population observation model, sampler, and random stream. Only the context component is replaced by the correct, wrong-context, repeatedly shuffled, population, or oracle operator. Correct calibration comes from disjoint support in the held-out query context; the population operator is estimated only from training contexts; and the oracle is restricted to semi-simulation. A zero gate is identical to the population-posterior condition and is not an unconditional null. The primary estimand is the paired correct-minus-population effect; correct-minus-wrong and correct-minus-shuffled effects are negative-control diagnostics, and oracle-minus-correct measures operator-estimation headroom. The gate requires [TBD: correct-versus-population effect, uncertainty, and preservation criteria]. Neither theory nor geometry guarantees that every wrong operator is worse on every metric. Failure removes the context-personalization claim.

RQ3: Does iterative diffusion add value beyond amortized inference? Two deterministic baselines separate supervision from inference. The *information-matched amortized baseline* sees only the clean prior-training records and the same synthetic partial-observation construction as the diffusion model; it receives the identical query, population and context operators, attenuation, gate, conditioning variables, and receptive field, and learns a one-pass conditional mean. The *strong task-matched baseline* is trained on available paired noisy-clean examples and may therefore receive more corruption supervision; it is reported as a conservative upper-strength comparator, not as an information-matched test. Architecture capacity, optimization budget, and exposure are tabulated rather than asserted identical when they cannot be. The diffusion decision rule is fixed before evaluation; no favorable sample is selected using its target. The gate passes only if diffusion supplies [TBD: preregistered advantage] in preservation, out-of-distribution robustness, decision utility, or empirically calibrated predictive information without violating latency and memory limits. Otherwise the iterative sampler is removed.

RQ4: Do learned attenuation and real-data restoration preserve neural structure? Only after RQ1–RQ3 pass do we replace the oracle event indicator with either an external auxiliary-sensor attenuation or a frozen EEG-only detector. The former is evaluated as a conditional likelihood input; the latter defines a data-dependent generalized-Bayes observation energy and is never described as an artifact-probability likelihood. Real recordings evaluate within-session calibration, cross-session transfer, and paradigm transfer separately. Artifact attenuation is paired with event-related potentials, spectral amplitude and phase, event-related desynchronization/synchronization, spatial covariance or topography, and fixed/retrained decoding where supported. The gate requires attenuation beyond [TBD: threshold] while every preregistered preservation outcome remains within [TBD: non-inferiority margins]. Semi-simulated fidelity and natural-recording preservation are never pooled. Failure restricts the result to the oracle mechanism.

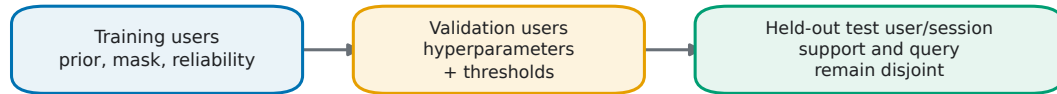
RQ5: Does the adaptive loop detect unreliable calibration and roll back to the population posterior? Controlled failures include insufficient ocular-regressor excitation, corrupted references, wrong-context support, channel loss, montage perturbation, support–query shift, and natural cross-session drift. The monitor is assessed by discrimination, false-trigger rate, detection delay, time in rollback, recovery after recalibration, and regret relative to the population posterior. Independently, a shared-seed numerical test sets the gate to zero and compares every reverse-sampler state with a direct population-posterior invocation. The implementation must short-circuit before any context precision, residual, Jacobian product, clipping, or context-dependent random-number consumption. The gate requires pathwise agreement within [TBD: numerical tolerance], failure-detection performance of [TBD: thresholds], and worst-context paired regret within [TBD: rollback margin]. Algebraic equality alone is insufficient: the detector must trigger when calibration is harmful, and the population route is not claimed harmless outside its validated distribution.

Deferred phase: Can a diagnosed primary failure motivate a frozen backup operator? Appendix E preserves six EEG-specific contingencies, but they are outside the first confirmatory campaign. The portfolio is opened only if P0 passes the oracle mechanism and specificity gates yet exhibits a preregistered, observable failure mode such as outlier-contaminated support, temporal mismatch, frequency heterogeneity, graph-stabilizable short support, covariance-only calibration, or low-information support with a compatible population bank. Activation requires a prospective protocol amendment that freezes one applicable family, its diagnostic, and its comparison before outer-test results are opened. No backup may rescue a failed P0 mechanism, and a post hoc best-family result is never deployable evidence.

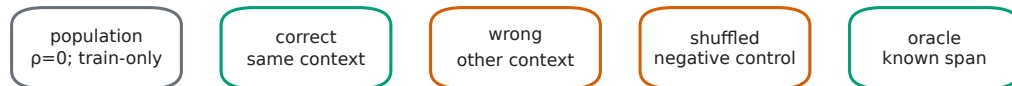
Table 2. Preregistered go/no-go gates. Thresholds are intentionally unset until the protocol is signed and remain independent of observed test results. A failed gate restricts the claim shown in the final column.

Gate	Confirmatory evidence	Required margin or operating limit	Consequence of failure
RQ1	Oracle-calibrated posterior versus population posterior and oracle-span subtraction	[TBD: span-specific effect, overlap, and complement margins]	Stop before learned components
RQ2	Correct versus population, wrong, shuffled, and oracle context operators	[TBD: specificity effect and uncertainty rule]	No context-personalization claim
RQ3	Diffusion versus information-matched and strong task-matched estimators	[TBD: utility, calibration, latency, and memory margins]	Replace diffusion with one-step inference
RQ4	Learned attenuation and real-data neural/task preservation	[TBD: attenuation and non-inferiority margins]	Oracle-mechanism claim only
RQ5	Failure detection, drift response, and exact population-posterior rollback	[TBD: detection, regret, recovery, and numerical limits]	No adaptive-deployment claim
Deferred	One prospectively activated backup after a diagnosed P0 failure	[TBD: amended family-specific criterion]	Retain validated P0 or refuse

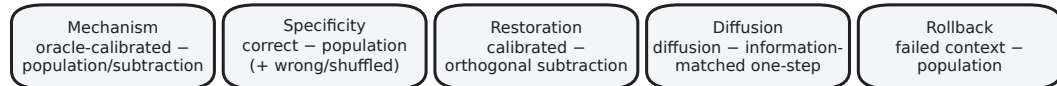
Grouped outer split



Calibration intervention



Decisive paired contrasts



Go / no-go evidence (participant or session is the inferential unit)



Fig. 4. Grouped, stage-gated evaluation. Dataset-specific population models and thresholds are frozen before a held-out context is opened. Operator-source swaps isolate mechanism and specificity; oracle-span subtraction and two one-step estimators test whether restoration and diffusion are necessary. The backup portfolio is deferred until a diagnosed P0 failure. No empirical value is shown.

6.2 Datasets and evidential roles

The Klados–Bamidis semi-simulated EEG/EOG resource is reserved for causal mechanism and paired waveform fidelity [23]. Its construction provides participant-dependent ocular propagation and a clean target, permitting oracle-projector, subtraction-versus-restoration, and subspace-specific error tests. The published sample extent and the number remaining after integrity checks will be recorded as [TBD: verified participants, sessions, and usable paired segments]. Mixtures are generated only after source records have been assigned to an outer fold, so neither a clean segment nor an ocular source can appear in both model development and held-out evaluation.

The real calibration study associated with SGEYESUB is the planned source for within-session ocular attenuation and preservation under instructed eye movements [25]. Access, reuse authorization, channel availability, and the auditable usable sample remain [TBD: data-access and cohort audit]. These recordings test the calibrated low-rank ocular premise and the difference between subspace subtraction and restoration, but they do not supply a latent clean waveform. If record-level access cannot be documented, this experiment is omitted rather than replaced by an undocumented reconstruction of the cohort.

The Eye-BCI multimodal, multi-paradigm resource is planned for cross-session and cross-paradigm validation because synchronized EEG, eye tracking, and task structure permit artifact timing to be separated from task-valid neural measurements [13, 14]. Its published and analyzable extents remain [TBD: verified participants, sessions, paradigms, and trials]. Within-session support/query evaluation and support-from-one-session/query-from-another evaluation are reported separately; the latter is a drift and transfer test, not evidence that the within-context stability assumption always holds.

EEGdenoiseNet is used only as a generic artifact stress test [47]. Its EOG mixtures can test whether a population restoration model transfers to a common benchmark, while its electromyographic (EMG) mixtures assess generic robustness to an out-of-scope corruption. EMG is not treated as a personalized low-rank ocular operator, no EMG result contributes to RQ2, and no success on this resource extends the paper’s personalized claim beyond ocular artifacts. Table 3 fixes these evidential boundaries. At the time of this draft, acquisition and integrity-ledger entries are [TBD: completion status for each resource]; every experiment lacking a signed ledger is described as planned.

Channel geometry is closed by training a separate clean prior and population observation model for every admitted dataset–montage–reference configuration. A result across datasets therefore tests procedural replication, not cross-montage checkpoint transfer. Channel count, order, reference, preprocessing, and population-operator-bank identifiers are part of each checkpoint key. No metadata token, implicit padding, or unvalidated interpolation is allowed to reconcile incompatible montages in the primary analysis.

6.3 Held-out contexts, support budgets, and leakage control

The outer split holds out every recording from a participant during model training, hyperparameter selection, reliability-gate fitting, threshold selection, and decoder development. Where a participant has multiple sessions, the manifest also defines session-grouped tests: within-session support and query are disjoint raw blocks, whereas cross-session tests take support and query from different sessions without allowing either session into population training. Training participants are subdivided into development and validation groups; no window from a held-out participant is used to choose rank, ridge strength, attenuation mapping, guidance schedule, number of sampler steps, checkpoint, or go/no-go margin.

Table 3. Dataset roles and admissible claims. “Paired” means that a construction supplies a clean target; it does not imply that a natural recording has a known artifact-free counterpart.

Resource	Primary role	Inferential unit and target	Explicit exclusion
Klados–Bamidis semi-simulated EEG/EOG	RQ1 mechanism; RQ2 oracle specificity; paired fidelity	Held-out participant; paired clean waveform and generating span	Not sufficient evidence of natural-recording physiology
SGEYESUB-associated calibration recordings	RQ4 within-session attenuation/preservation; subtraction control	Held-out participant/session; no latent clean target	No claim until access and reuse are verified
Eye-BCI multimodal multi-paradigm data	RQ4 cross-session/paradigm preservation; RQ5 drift	Held-out participant/session; task-valid physiological outcomes	No pooled score with semi-simulation
EEGdenoiseNet	External EOG and generic EMG stress tests	Source-record split; paired mixtures where defined by the benchmark	No personalized EMG or context-operator claim

The common support grid is exactly 0, 5, 10, 30, 60, or 120 seconds, but a dataset contributes only the budgets supported by its native, disjoint calibration material. In particular, a native Klados–Bamidis recording contains only 30 seconds: 60- and 120-second support must never be manufactured by repeating, overlapping, or resampling one recording. Those two budgets are admissible only if a preregistered rule concatenates distinct calibration blocks from multiple recordings belonging to the same held-out context while keeping every contributing block disjoint from the query; otherwise they are marked unavailable. The zero-second condition supplies no context operator and enforces $\rho = 0$; it executes the training-only population observation posterior on the same query, not an unconditional prior or zero-filled calibration input. Nonzero supports are selected by a prespecified chronological or seeded rule, not by retrospective selection of the clearest eye movements. Coverage of horizontal, vertical, blink, and eyelid-related regressor excitation is recorded as a reliability diagnostic rather than silently guaranteed. Each support block, query block, trial identifier, time range, and guard interval is stored in an immutable manifest. No raw sample, overlapping window, event, or filter transient crosses the support–query boundary.

Raw recordings are assigned to groups and blocks before any operation that could share state. Channel mapping, referencing, filtering, resampling, artifact annotation, normalization, and segmentation are then applied independently according to the configuration in Appendix C. Normalization parameters and learned preprocessing are estimated from population-training data only unless a method explicitly permits support-only corruption calibration; query statistics are never used. Query precision attenuation is either supplied by a declared external modality or produced by an EEG-only detector frozen on training contexts; the latter is evaluated as generalized-Bayes guidance. Support estimates only the transfer operator and its reliability diagnostics: it never updates the population prior, a participant embedding, a downstream decoder, or a test-time hyperparameter.

All stochastic training and sampling is repeated with [TBD: number and identities of preregistered seeds]. Seeds, shuffled pairings, mixture realizations, and support selections are common across paired method comparisons. A seed or a window is never counted as an independent participant. Results are first aggregated across windows and stochastic repetitions within a participant/session; the resulting context-level values enter inference.

6.4 Primary operator and deferred portfolio

The first campaign implements only P0, the instantaneous ridge estimate of the EOG-reference transfer followed by a training-frozen top- r SVD. It also implements a training-only population operator, the zero-gate population-posterior route, wrong/shuffled controls, and a semi-simulated oracle. These are sufficient to identify whether the proposed mechanism and context specificity exist. Robust PCA and post hoc mixtures are excluded because generic low-rank-plus-sparse structure does not identify ocular activity from neural EEG.

Appendix E records six possible responses to distinct, observable P0 failures: block-robust regression, lag expansion, frequency-selective precision, graph regularization, covariance contrast, and population shrinkage. The appendix is a design reserve, not a simultaneous model-selection policy. A family can enter code-to-result execution only after the primary gates pass and a prospective amendment freezes its triggering diagnostic, support-only validation, hyperparameters, compute cap, and new outer-test split. A valid P0 is never displaced by a higher support score, and query clean targets never select a family. Query-time monitoring may reduce ρ or request a new support episode; it cannot switch families in place.

Within the primary experiments, correct, population, wrong, shuffled, and oracle conditions reuse the identical dataset-specific prior checkpoint, query attenuation, population energy, guidance schedule, sampler decision rule, and random stream. Only the context operator changes. The semi-simulated clean target is opened only after ranks, gates, and operator sources are frozen. Klados–Bamidis, SGEYESUB-associated, and Eye-BCI cells are enabled only when their audited sensors, event labels, durations, and electrode metadata satisfy P0’s assumptions. EEGdenoiseNet cannot identify a personalized multichannel transfer operator and remains a population stress test.

6.5 Baselines and controlled interventions

The classical comparison includes the raw observation, common standard preprocessing, forward and backward EOG regression, ICA with a frozen automatic component-labeling rule, Artifact Subspace Reconstruction (ASR), a multichannel Wiener filter, and the native SGEYESUB correction. ASR is treated as deterministic covariance-based reconstruction, not as a method that merely deletes a subspace and not as a probabilistic posterior. The native sequential SGEYESUB correction is implemented according to its audited reference code and is kept distinct from the synthetic-control comparator called *oracle-span orthogonal subtraction*, which applies $Q = I - \Pi$. Each method receives the channels and calibration information assumed by its published protocol. ICA duration, component rule, ASR calibration and cutoff, regression direction and regularization, and subspace rank are fixed on training-context validation data.

The deterministic learned comparison includes CLEEGN, EEGOAR-Net, A2DM, a compact band-routing denoiser, and a strong multichannel convolutional or U-Net estimator [26–29]. A2DM is artifact-aware but is not categorized as diffusion. The diffusion comparison includes EEGDfus, D4PM, and other directly applicable diffusion implementations for which a faithful configuration can be audited [17, 35]. The population-posterior condition uses the identical clean prior, training-only population observation model, query, sampler, and seeds but no test-context correction. The information-matched one-step estimator uses the same clean records and synthetic partial observations as diffusion; the strong task-matched estimator may additionally use paired noisy-clean data. Parameter count, updates, accelerator time, paired-supervision exposure, and corruption exposure are reported so capacity, supervision, personalization, and iterative inference are not conflated.

Within CSPD, the operator-source intervention substitutes oracle, correct, wrong-context, repeatedly shuffled, or population calibration while holding the trained prior, population energy, query, attenuation, sampler, and random

stream fixed. The population condition is implemented by an explicit $\rho = 0$ short-circuit and verified against a direct population-posterior invocation; there is no separate unconditional “null” comparator in the primary estimand. The subtraction-versus-restoration intervention applies the same oracle span and hard event timing either as exact orthogonal deletion or as a context observation energy. Attenuation ablations compare oracle hard events, external-sensor attenuation, EEG-only attenuation, always-full precision, and always-attenuated precision. Gate ablations compare the learned fixed-per-window reliability value with always-context, individual diagnostic removals, and hard rollback. Sensitivity experiments vary estimated rank, ridge, attenuation calibration, observation variance, guidance schedule, sampler family and step count, support budget, and reliability threshold over a grid frozen before test evaluation.

6.6 Outcomes and operating characteristics

Paired semi-simulated outcomes quantify full-waveform and subspace-resolved fidelity. They include relative root-mean-square error, temporal correlation, signal-to-noise improvement, spectral error, errors inside the oracle ocular span and its complement, and $r_{\text{overlap}} = \|\Pi X\|_F^2 / \|X\|_F^2$. The reported diffusion decision is [TBD: sample mean or other prespecified rule and sample count]; a favorable individual restoration sample is never selected after observing its target. RQ1 emphasizes recovery of the neural component inside the ocular span, whereas complement error tests whether improvement is purchased by excessive deviation from the measurement-constrained directions. Finite variance means exact equality is neither assumed nor required.

Primary-operator outcomes include P0 eligibility, transfer-regressor excitation, retained rank and singular gap, bootstrap principal angles, false acceptance of harmful context calibration, false refusal of beneficial calibration, population-route rate, and regret relative to the population and oracle conditions. If the deferred portfolio is later activated, family eligibility, attempted fits, selection, cost, and failure are reported in a separately amended analysis. Such diagnostics cannot substitute for waveform fidelity or physiological preservation.

Natural-recording artifact outcomes include blink- and saccade-locked residual amplitude, EEG–EOG coherence, frontal residual topography, and eye-tracking-locked contamination where synchronized eye tracking is available. These are paired with neural-preservation outcomes appropriate to each paradigm: clean-interval waveform change; band-power and phase distortion; event-related-potential amplitude, latency, and scalp distribution; steady-state response amplitude and phase; motor-imagery desynchronization/synchronization; and covariance or connectivity summaries whose estimators are fixed in advance. An outcome is omitted when a dataset does not support its physiological interpretation. Artifact attenuation alone cannot establish acceptable cleaning because a zero signal can score well on residual ocular energy.

Downstream evaluation uses both fixed and retrained decoders. The fixed decoder is trained once on the training-context version of raw or standard-preprocessed EEG and then receives each cleaning condition, measuring whether preprocessing preserves the original decision rule and calibration. The retrained decoder uses the same architecture, optimization, trial aggregation, and held-out-context split but is retrained for each cleaning condition, measuring whether task information remains recoverable after the input distribution changes. Raw and standard-preprocessed controls appear in both analyses. Accuracy or balanced accuracy, probabilistic calibration, and task-appropriate discrimination are reported at the participant/session level; decoder specifications and checkpoint rules are frozen before test inference.

Predictive evaluation treats restoration samples as model outputs, not physiological certificates or exact posterior draws. Paired data support coordinate-wise and joint-summary nominal coverage, interval width, proper interval scores, sample-mean error, and calibration error at [TBD: preregistered levels]. Only after held-out coverage is demonstrated

are the intervals called empirically calibrated predictive intervals. Natural recordings support sampler dispersion in attenuated directions, stability in measurement-constrained directions, uncertainty-based refusal, and consistency with physiological events. Error-uncertainty correlation is only a ranking diagnostic. Operator uncertainty is distinguished from waveform variation; with a plug-in projector, all coverage statements are conditional on that estimate.

Deployment measurements include support-processing wall time, calibration energy where measurable, CPU and GPU inference latency, throughput, peak host and accelerator memory, parameter and checkpoint size, and quality-latency curves over reduced sampler steps. Latency distributions include tail rather than mean latency alone, with hardware, batch size, precision, warm-up, and power-measurement method fixed. The online boundary is the preregistered limit [TBD: window deadline, throughput, memory, and energy budget]; a method exceeding it is described as offline even if its fidelity gate passes.

Drift experiments replay the adaptive loop under natural session changes and controlled shifts in channel availability, reference quality, montage mapping, support excitation, and ocular propagation. They report reliability trajectory, false triggers, detection delay, time to fallback, fallback duration, and performance before and after recalibration. Privacy evaluation treats the stored projector, reliability features, and support summaries as potentially linkable biometric-adjacent data. It measures [TBD: preregistered membership, linkage, or re-identification attack and threat-model metric] against raw-support and population-only references, while recording storage duration, byte footprint, transmission, and deletion behavior. This is a risk audit, not a claim that a small matrix is anonymous.

6.7 Statistical analysis and multiplicity

Every confirmatory comparison is paired within the same held-out participant/session, support draw, query set, mixture realization, and stochastic seed. Window-level metrics are summarized within context before inference. When several sessions belong to one participant, the participant is the outer resampling cluster and sessions are resampled or aggregated within participant according to the preregistered estimand. Shuffled pairings and sampler seeds are averaged within context and also summarized as algorithmic variability; they do not inflate the biological sample size.

For each primary contrast, we report the paired effect in natural units, a standardized or rank-based paired effect size, and a two-sided [TBD: confidence level] interval from a participant-level paired bootstrap. The bootstrap resamples paired contexts, preserving all method labels, and uses [TBD: number of resamples and interval construction]. Hypothesis tests use an exact or Monte Carlo paired sign-flip/permutation test when its exchangeability condition is defensible; otherwise the prespecified alternative is a two-sided Wilcoxon signed-rank test, with ties and zero differences handled by a declared rule. Distributional diagnostics accompany either choice. The five RQ-level confirmatory p -values are adjusted by Holm's step-down procedure. Any later, prospectively amended backup-family comparisons form their own Holm-adjusted family and cannot alter the original five-gate decisions. Secondary outcomes are grouped into fidelity, preservation, predictive calibration, deployment, and rollback families with separately declared Holm corrections; exploratory analyses are labeled and are not used to rescue a failed primary gate.

Missing recordings, method failures, and abstentions are outcomes rather than silently discarded windows. The analysis reports the attempted and analyzable participant/session counts as [TBD: flow counts by dataset and method], reasons for exclusion, and a paired complete-case analysis alongside [TBD: prespecified robustness analysis for method failures]. No claim is based solely on statistical significance: the confidence interval must also cross the applicable practical, non-inferiority, latency, privacy, or rollback margin in the direction required by Table 2.

6.8 Reproducibility and reporting boundary

Appendix C defines the dataset ledger, immutable split manifests, preprocessing and baseline provenance, environment and seed records, and one command per output. Appendix D defines metrics, inferential units, empty result tables, and machine-readable figure-data schemas. A result can enter the manuscript only when its record binds the source commit, environment, dataset and split hashes, configuration, checkpoint, seeds, command, and output checksum. Until those contracts are populated from the clean-room implementation, every result cell is an explicit [TBD: estimate, uncertainty interval, sample extent, and provenance identifier], and no placeholder is evidence.

7 Limitations, Ethics, and Responsible Adaptation

7.1 Model and evidential limits

The low-rank observation model is deliberately narrow. It is plausible for the spatial propagation of ocular sources over a stable user-session-montage context, but it does not state that clean EEG is low rank, that all eye-related contamination lies in the calibrated span, or that the reliable complement is noise-free. Eyelid motion, electrode motion, saturation, reference contamination, and changing contact impedence can remain in the mismatch term. A short support recording can also be systematically biased when instructed eye movements co-occur with neural responses or fail to excite every relevant ocular direction. Ridge regularization improves conditioning but does not make such support representative. The empirical protocol must therefore report excitation, conditioning, bootstrap stability, and support-query shift rather than treating the word “calibration” as evidence of identifiability.

The deferred backup portfolio would not remove this identifiability problem. Robust loss can reject useful neural blocks, a lag dictionary can consume the sensor space, a filter-bank covariance can confuse low-frequency neural activity with ocular variance, graph regularization can impose an incorrect scalp smoothness pattern, and CCA/GEVD can select neural responses correlated with prompted eye movements. Population shrinkage can conceal a genuine context shift. These are possible follow-up designs, not evidence that one family must succeed and not routes in the current confirmatory campaign. A future family-specific protocol must retain the population-posterior rollback even when a post hoc operator comparison looks favorable.

Within-context stability is an assumption, not a permanent user trait. Removing and replacing a cap, changing the reference or montage, losing a channel, or moving to a later session can invalidate a previously reliable operator. The proposed monitor can request recalibration, but frequent recalibration increases burden and may be incompatible with passive or inaccessible interfaces. Synchronized EOG or eye tracking during support is another deployment requirement, and an external query-time attenuation trace may require sensors unavailable in a minimal headset. Experiments must distinguish external-sensor and EEG-only attenuation and must not present a laboratory calibration protocol as calibration-free or zero-shot operation.

The personalized claim is ocular-only. EMG, motion, line noise, and electrode transients have different spatial and temporal structure and need not be captured by a short low-rank ocular calibration. EEGdenoiseNet EMG experiments are consequently generic stress tests of the population denoiser, not evidence for personalized EMG removal. A failure on EMG does not falsify the ocular operator model, and success does not broaden it.

The first implementation trains a separate clean-EEG prior and population observation model for each admitted dataset and montage. Conditioning on montage metadata cannot by itself reconcile different channel counts, orders, references, or missing sensors. Accordingly, cross-dataset results test whether the same procedure replicates under

separately trained priors; they do not demonstrate that one checkpoint generalizes across montages. A future canonical-montage transform or variable-channel prior would require separate analysis of rank, noise covariance, interpolation error, and missing-channel uncertainty.

The evidential regimes also answer different questions. Semi-simulated mixtures permit paired fidelity and an oracle intervention but inherit the source-cleaning and mixture assumptions used to construct them. Natural recordings preserve realistic physiology but lack a counterfactual artifact-free waveform. Artifact attenuation, task preservation, and cross-session consistency jointly constrain the interpretation; none proves that the reconstructed signal is the one that would have occurred without the ocular event. Dataset sizes, exclusions, access conditions, and demographic or recording diversity remain [TBD: verified dataset and cohort audit]. Claims will be narrowed if the available montages, paradigms, hardware, or participant populations do not support broader generalization.

7.2 Predictive uncertainty and rollback limits

Model-based restoration introduces a failure mode absent from transparent subtraction: a plausible clean-looking waveform may be generated where the observation contains little information. Samples from the implemented finite-step, plug-in sampler represent model-induced variation, not exact posterior draws or proof of physiological truth. Measurement-constrained directions remain affected by finite observation variance, guidance tempering, and clipping, while attenuated directions depend strongly on the population prior and can be hallucinated or over-smoothed under misspecification. This risk is most important when downstream users interpret morphology, phase, or transient activity rather than aggregate task performance. The system must expose sample dispersion and its calibration status, preserve the raw observation and provenance, and abstain when uncertainty or support diagnostics exceed preregistered limits. It must never silently replace the sole clinical or scientific record with a generated waveform.

The initial implementation conditions on a plug-in projector or precision operator. Its nominal predictive intervals therefore omit calibration-operator and family-selection uncertainty and cannot be advertised as Bayesian credible intervals for the cleaned waveform. Bootstrap subspace or precision variability can inform the reliability gate, but it is not equivalent to marginalizing over the operator or selector. Only intervals whose coverage is verified on held-out paired semi-simulation may be called empirically calibrated predictive intervals, and that calibration may fail after a montage or population shift. Any deployment must therefore monitor coverage proxies and preservation diagnostics without interpreting low sampler variance as evidence that the model is correct.

The exact fallback theorem is algebraic. It guarantees pathwise equality to the population-posterior sampler only when a zero gate takes an explicit short-circuit before constructing any context operator, residual, Jacobian product, or calibrated guidance, and both executions share the same randomness. It does not guarantee that the learned detector recognizes every harmful calibration, that a small positive gate causes small downstream risk, or that the population observation model is harmless. The detector, trigger, rollback, and recovery behavior are separate empirical obligations under RQ5. If they fail, the appropriate response is to disable context adaptation; it is not to lower the reporting threshold. The population-posterior route may itself need to abstain when the query or montage is out of distribution.

Iterative sampling also incurs time, memory, and energy costs. Reduced-step inference can alter fidelity and calibration, so an “online” claim depends on the measured tail latency and throughput on declared hardware, not on the existence of a fast sampler in principle. If CSPD does not satisfy [TBD: preregistered operating limits], it is restricted to offline analysis. If a matched one-step estimator reaches equivalent preservation and decision quality without a useful uncertainty loss, the responsible design is to remove diffusion and its additional computational burden.

7.3 Human data, privacy, and foreseeable misuse

EEG, EOG, eye-tracking records, and calibration-derived operators are potentially sensitive even when a public repository distributes de-identified files. A projector, covariance, graph-smoothed map, or population-alignment coefficient can encode stable aspects of anatomy, cap placement, and acquisition practice and may permit cross-session linkage. Reliability features, selector traces, and support summaries can reveal recording quality or behavior. The threat model therefore includes unauthorized linkage, membership inference, re-identification through auxiliary data, and reconstruction of calibration behavior. The planned privacy evaluation does not make these objects anonymous; it estimates risk under explicitly stated attacker knowledge.

The preferred deployment keeps raw support, the projector, and diagnostics local to the acquisition device; encrypts them at rest and in transit when transmission is unavoidable; applies least-privilege access; separates identity from experimental identifiers; records every export; and deletes context state when the session ends or the participant withdraws, subject to an approved retention policy. A stale projector should be deleted rather than accumulated as a longitudinal biometric profile. Exact storage location, retention duration, access roles, deletion verification, and incident response remain [TBD: approved data-governance plan]. The raw signal must not be discarded merely because a cleaned derivative exists.

For every reused dataset, the final paper must report the original consent and ethics statement, license or data-use agreement, permitted secondary use, and any applicable local determination. Public availability alone is not an ethics exemption. Any newly collected calibration data require approval before acquisition, informed consent, compensation and withdrawal terms where applicable, and a protocol identifier; these fields remain [TBD: ethics body, determination or approval identifier, consent procedure, and collection status]. Analyses that could enable identity inference or disclose sensitive attributes require specific review rather than being treated as ordinary denoising benchmarks.

Foreseeable harms include suppressing genuine neural events, inserting population-typical structure, changing a brain-computer-interface decision, concealing sensor failure, and encouraging clinical interpretation of a research model. The present work is not a diagnostic device and does not authorize unattended high-stakes use. Cleaned outputs must carry the method version, calibration and gate status, uncertainty summary, fallback events, and links to the raw record. Human operators need a refusal mode and a way to inspect why recalibration was requested.

Performance can vary with anatomy, eye-movement ability, disability, hair and skin properties that affect electrode contact, task strategy, and equipment. Reporting only a global mean may hide systematic failure. Where consent and sample support allow, subgroup and worst-context analyses will be reported with uncertainty and privacy-preserving aggregation; categories and minimum reporting sizes remain [TBD: ethically justified subgroup plan]. Such analysis is diagnostic rather than a claim that observed differences are biological.

7.4 Accountability and research-process disclosure

All authors remain responsible for dataset authorization, code, derivations, citations, outputs, and claims. The clean-room process has used OpenAI Codex for research-lifecycle tasks beyond spelling correction, including structuring the method and experiments, checking derivations, and drafting code and prose. Appendix C therefore reserves a detailed audit of the tool version and access date, inputs, retained outputs, prohibited uses, and independent human verification. This disclosure must be reconciled with the ACM policy in force on the submission date. No AI system is an author, no generated citation or empirical value is accepted without primary-source or provenance verification, and every human author must review the entire submitted artifact rather than only an assigned section.

Finally, this document is not a submission while it contains [TBD: empirical, ethics, authorship, and archival fields]. Funding, conflicts of interest, author contributions, data custodians, and any editorial approval required for the reconceived manuscript must be completed through the journal’s process. Passing the staged scientific gates cannot substitute for those obligations.

8 Conclusion

This work reframes unseen-user ocular artifact removal as calibration of a changing corruption mechanism rather than adaptation of a person’s neural representation. CSPD estimates only a context-specific transfer from specified ocular reference regressors to EEG sensors, using support that is strictly disjoint from deployment queries. A dataset-specific clean-EEG prior is paired with a training-only population observation energy and a support-calibrated context energy, both conditioned on the same query. Reliability interpolates between those energies. Consequently, a zero gate is an executable rollback to population-posterior denoising, not an unconditional generation path, while a positive gate applies only the supported likelihood-ratio correction. The context energy measurement-constrains the estimated complement under finite variance and attenuates precision in the ocular span; it does not claim to fix any coordinate exactly.

The contribution is conditional on evidence, not on the use of diffusion itself. The staged protocol first asks whether oracle-calibrated restoration improves the ocular-span component over oracle-span orthogonal subtraction without degrading the measured complement, and reports how much clean-signal energy actually overlaps that span. It then asks whether correct context calibration improves on the population observation model and behaves differently from wrong and shuffled controls. Diffusion is retained only if it adds preservation, out-of-distribution robustness, decision utility, or calibrated predictive information beyond an information-matched amortized estimator; a separately reported paired noisy-clean estimator supplies a strong task-matched comparator. Learned attenuation, drift detection, and rollback are introduced only after these mechanism tests. The six backup operators remain a deferred contingency plan, activated prospectively only when the primary ridge/SVD operator is scientifically useful yet fails in a diagnosed way.

This clean-room manuscript currently reports no empirical outcome. Every sample extent, effect, interval, operating cost, privacy result, and gate decision remains [TBD: generated value with a complete provenance record]. If the gates pass, the evidence would support a limited conclusion: ocular corruption-only calibration can adjust a population observation posterior toward the current context while retaining an auditable rollback. It would not show that model-based restoration samples are physiological ground truth, that nominal intervals are calibrated without coverage tests, that the method is zero-shot, that one calibration remains valid across sessions, that one prior spans incompatible montages, or that the low-rank ocular model extends to EMG and other artifacts.

The most important negative result is specified in advance. If the information-matched one-step estimator reaches equivalent preservation and downstream utility and diffusion does not provide decision-relevant calibrated predictive information, the iterative component should be removed. Likewise, failure of correct-versus-population specificity removes the personalization claim, failure of natural-recording preservation restricts the result to mechanism evidence, and failure of the detector disables context adaptation. A failed primary mechanism cannot be rescued post hoc by searching the deferred portfolio. In that sense, “personalize the corruption, not the brain” is both a modeling principle and a claim boundary: adapt only what can be identified from support, expose what remains uncertain, and roll back to the population observation model when the context cannot be trusted.

Acknowledgments

The final manuscript must identify funding, data custodians, and any human-subjects approvals applicable to newly conducted analyses. During clean-room manuscript development, OpenAI Codex was used to help structure prose, check symbolic derivations, draft plotting code, and query public bibliographic metadata. Human authors must independently verify every statement, citation, proof, figure, and experimental value and remain solely accountable for the submitted work. This statement must be reconciled with the ACM artificial-intelligence policy in force on the submission date.

A Complete Proofs and Auxiliary Results

This appendix derives the analytical statements used in the main text. The results concern the retained EOG-reference transfer model and the specified energy update. They are not distribution-free guarantees of physiological recovery.

A.1 Retained-space ridge calibration

PROPOSITION A.1 (RIDGE ERROR IDENTITY). *Let $Y_c = C_R \tilde{E}_c + N_c$, $G_R = \tilde{E}_c \tilde{E}_c^\top \succ 0$, and $D_\lambda = G_R + \lambda I_q$. The minimizer of $\|Y_c - D \tilde{E}_c\|_F^2 + \lambda \|D\|_F^2$ over D is $\hat{C}_{\lambda,R} = Y_c \tilde{E}_c^\top D_\lambda^{-1}$ and satisfies Equation (30) with $C_{\omega,R} = C_R$. Moreover,*

$$\|\hat{C}_{\lambda,R} - C_R\|_2 \leq \frac{\|N_c \tilde{E}_c^\top\|_2 + \lambda \|C_R\|_2}{\lambda_{\min}(G_R) + \lambda}. \quad (48)$$

PROOF. Differentiating the objective gives $2(D \tilde{E}_c - Y_c) \tilde{E}_c^\top + 2\lambda D = 0$ and hence $DD_\lambda = Y_c \tilde{E}_c^\top$. Substituting the support model yields

$$\hat{C}_{\lambda,R} = C_R G_R D_\lambda^{-1} + N_c \tilde{E}_c^\top D_\lambda^{-1}.$$

Because $G_R D_\lambda^{-1} = I_q - \lambda D_\lambda^{-1}$, subtracting C_R proves Equation (30). Submultiplicativity and $\|D_\lambda^{-1}\|_2 = (\lambda_{\min}(G_R) + \lambda)^{-1}$ prove Equation (48). $\square$

The restriction to $\mathcal{R}_E = \text{range}(V_R)$ is essential. If the full Gram matrix $E_c E_c^\top$ is singular, two transfers that agree on $\text{range}(E_c)$ produce the same fitted support signal. No estimator can distinguish their action on $\ker(E_c^\top)$ without additional assumptions. In contrast, $G_R \succ 0$ makes the transfer from the retained coordinates unique under the noiseless linear model.

For completeness, suppose only in this paragraph that, conditional on $\tilde{E}_c$, entries of N_c are independent, zero mean, and have variance σ_N^2 . Equation (30) then gives

$$\mathbb{E}[\hat{C}_{\lambda,R} - C_R \mid \tilde{E}_c] = -\lambda C_R D_\lambda^{-1}, \quad (49)$$

$$\mathbb{E}[\|\hat{C}_{\lambda,R} - C_R\|_F^2 \mid \tilde{E}_c] = \lambda^2 \|C_R D_\lambda^{-1}\|_F^2 + C \sigma_N^2 \text{tr}(G_R D_\lambda^{-2}). \quad (50)$$

For a row n of N_c , $\mathbb{E}[\|n \tilde{E}_c^\top D_\lambda^{-1}\|_2^2] = \sigma_N^2 \text{tr}(G_R D_\lambda^{-2})$; summing over EEG channels and using the zero conditional mean removes the cross term. Prompted eye movements can violate this assumption through neural-EOG correlation, so the deterministic identity is the primary statement.

A.2 Top-r truncated SVD and Wedin perturbation

PROPOSITION A.2 (TOP-R WEDIN BOUND). *Let $A \in \mathbb{R}^{C \times q}$ and $\hat{A} = A + Z$. Let $\Pi_r(A)$ and $\Pi_r(\hat{A})$ denote the orthogonal projectors onto their leading r left singular subspaces. Define $\Delta_r = \sigma_r(A) - \sigma_{r+1}(A) > 0$ and $e = \|Z\|_2$. If $e < \Delta_r$, then*

$$\|\Pi_r(\hat{A}) - \Pi_r(A)\|_2 \leq \min\left\{1, \frac{\sqrt{2}e}{\Delta_r - e}\right\}. \quad (51)$$

PROOF. Write the leading singular triplets of $\hat{A}$ as $(\hat{U}_1, \hat{\Sigma}_1, \hat{V}_1)$ and let (U_2, V_2) span the unselected left and right singular subspaces of A . Weyl's inequality gives $\hat{\sigma}_r \geq \sigma_r(A) - e$, so the selected singular values of $\hat{A}$ are separated from the unselected singular values of A by at least $\Delta_r - e$. Applying the same argument with A and $\hat{A}$ interchanged gives the second separation required by Wedin's coupled sin- Θ inequality.

The two residuals in that inequality are

$$R = A\hat{V}_1 - \hat{U}_1\hat{\Sigma}_1 = -Z\hat{V}_1, \quad S = A^\top\hat{U}_1 - \hat{V}_1\hat{\Sigma}_1 = -Z^\top\hat{U}_1,$$

and therefore $\|R\|_2, \|S\|_2 \leq e$. Expanding these residual equations in the singular-vector bases of A produces a pair of Sylvester equations for $U_2^\top\hat{U}_1$ and $V_2^\top\hat{V}_1$. The coupled sin- Θ residual inequality bounds either solution by the joint residual divided by the minimum singular-value separation:

$$\max\left\{\|U_2^\top\hat{U}_1\|_2, \|V_2^\top\hat{V}_1\|_2\right\} \leq \frac{(\|R\|_2^2 + \|S\|_2^2)^{1/2}}{\Delta_r - e} \leq \frac{\sqrt{2}e}{\Delta_r - e}.$$

Thus $\|U_2^\top\hat{U}_1\|_2 \leq \sqrt{2}e/(\Delta_r - e)$. For equal-dimensional subspaces this sine of the largest principal angle equals $\|\Pi_r(\hat{A}) - \Pi_r(A)\|_2$. Projector distance is at most one, which supplies the minimum. $\square$

Applying the proposition with $A = A_\lambda$ and $Z = Z_\lambda$ from Equation (31) proves Equation (33). Because $G_R \succ 0$, every eigenvalue of $R_\lambda = G_R(G_R + \lambda I)^{-1}$ is positive, so R_λ is invertible. If $C_{\omega,R}$ has rank exactly r , right multiplication by R_λ preserves its complete column range and hence $\Pi_r(A_\lambda) = \Pi(C_{\omega,R})$. If $C_{\omega,R}$ has rank greater than r , invertibility preserves the complete range but need not preserve the ordering or orientation of its leading r left singular vectors. The triangle inequality then proves Equation (34); the second term is a genuine target shift, not sampling noise.

A.3 Exact projector and numerical shrinkage

Let $C = U\Sigma V^\top$ be a thin rank- r SVD with $\sigma_1 \geq \dots \geq \sigma_r > 0$. Direct substitution gives

$$\Pi(C) = UU^\top, \quad S_\varepsilon(C) = U \operatorname{diag}\left(\frac{\sigma_i^2}{\sigma_i^2 + \varepsilon}\right) U^\top.$$

The two matrices are simultaneously diagonal on $\operatorname{range}(C)$ and zero on its complement. Hence $0 \preceq S_\varepsilon \preceq \Pi \preceq I$, and the maximum eigenvalue of $\Pi - S_\varepsilon$ is $\varepsilon/(\sigma_r^2 + \varepsilon)$. If $b_i = \sigma_i^2/(\sigma_i^2 + \varepsilon)$, then

$$|b_i^2 - b_i| = \frac{\varepsilon\sigma_i^2}{(\sigma_i^2 + \varepsilon)^2},$$

which proves Equation (35). Finally,

$$(I - S_\varepsilon)C = U \operatorname{diag}\left(\frac{\varepsilon\sigma_i}{\sigma_i^2 + \varepsilon}\right) V^\top \neq 0.$$

Thus even a perfectly known transfer does not yield exact cancellation under the shrinkage matrix.

A.4 Complement, subtraction, and estimated-projector bounds

PROOF OF THEOREM 5.1. Every column of $C_\omega E$ lies in $\text{range}(C_\omega)$, so $\Pi C_\omega E = C_\omega E$ and $Q C_\omega E = 0$. Multiplying Equation (2) by Q proves Equation (36). $\square$

For subtraction, $QY - X = -\Pi X + Q\Xi$. At each time point the two terms lie in orthogonal spatial subspaces; their Frobenius inner product is therefore zero. Expanding the squared norm proves Equation (37).

For an estimated projector, use $\Pi C_\omega E = C_\omega E$ to obtain

$$(I - \widehat{\Pi})C_\omega E = (\Pi - \widehat{\Pi})C_\omega E.$$

Submultiplicativity proves Equation (39). Likewise,

$$(I - \widehat{\Pi})X - QX = (\Pi - \widehat{\Pi})X,$$

which proves Equation (40). If $\widehat{C}_r$ is the rank- r truncation used for $\widehat{\Pi}$, $\widehat{S}_\epsilon = S_\epsilon(\widehat{C}_r)$, and $\widehat{\sigma}_r$ is the smallest retained singular value, then

$$\|\widehat{S}_\epsilon - \Pi\|_2 \leq \|\widehat{S}_\epsilon - \widehat{\Pi}\|_2 + \|\widehat{\Pi} - \Pi\|_2 = \frac{\epsilon}{\widehat{\sigma}_r^2 + \epsilon} + \delta_\Pi.$$

Repeating the preceding two arguments proves the stated shrinkage-adjusted bounds.

A.5 Reliability-amplitude precision

Let U span Π and $Q = I - \Pi$. For an externally supplied nuisance variance $\kappa_\tau^2 \geq 0$, set $a_\tau = \{\sigma_y^2 / (\sigma_y^2 + \kappa_\tau^2)\}^{1/2} \in (0, 1]$ as in Equation (18), and consider

$$y_\tau = x_\tau + U h_\tau + \xi_\tau, \quad h_\tau \mid a_\tau \sim \mathcal{N}(0, \kappa_\tau^2 I_r) = \mathcal{N}(0, \sigma_y^2 (a_\tau^{-2} - 1) I_r), \quad \xi_\tau \sim \mathcal{N}(0, \sigma_y^2 I_C).$$

Marginalizing h_τ yields

$$\Sigma_\tau = \sigma_y^2 Q + \sigma_y^2 a_\tau^{-2} \Pi, \quad \Sigma_\tau^{-1} = \sigma_y^{-2} (Q + a_\tau^2 \Pi). \quad (52)$$

Because $Q\Pi = 0$ and both are idempotent, $M_\tau = Q + a_\tau^2 \Pi$ satisfies $M_\tau^\top M_\tau = Q + a_\tau^2 \Pi$, proving Equations (15) and (41). The full Gaussian negative log-likelihood also contains

$$\frac{1}{2} \log \det \Sigma_\tau = \frac{C}{2} \log \sigma_y^2 - r \log a_\tau.$$

It is constant in x when an external a_τ is supplied, but it cannot be omitted when a_τ or a variance is fitted by the same likelihood.

As $a_\tau \downarrow 0$, variance in $\text{range}(\Pi)$ diverges. With an orthogonal basis $[U_\perp, U]$, the finite observation is

$$U_\perp^\top y_\tau = U_\perp^\top x_\tau + \epsilon_\tau, \quad \epsilon_\tau \sim \mathcal{N}(0, \sigma_y^2 I_{C-r}),$$

which has an ordinary density on the observed complement. This is a partial observation, not a singular full-dimensional density.

In the EEG-only setting $a_\tau = h_\phi(y)_\tau$. Although a_τ is held constant when differentiating with respect to x , the integral

$$\int \exp[-\mathcal{E}(x; y, h_\phi(y))], dy$$

generally depends on x . Consequently, dividing by a constant independent of x does not in general normalize the factor as a likelihood over y . Multiplication by a proper prior and normalization over x instead defines the generalized-Bayes target in Equation (20).

For the lifted block-diagonal operator $\mathcal{W}_j = \text{blockdiag}(w_\tau W_{j,\tau} / \sigma_y^2)_{\tau=1}^T$,

$$\mathcal{E}_j(x; y) = \frac{1}{2}(x - y)^\top \mathcal{W}_j(x - y), \quad \nabla_x \mathcal{E}_j = \mathcal{W}_j(x - y).$$

Linearity proves $\mathcal{W}_\rho = (1 - \rho)\mathcal{W}_0 + \rho\mathcal{W}_C$, Equation (24), and the following proposition.

PROOF OF PROPOSITION 5.2. For symmetric $W \succeq 0$,

$$d\{\frac{1}{2}(x - y)^\top W(x - y)\} = (x - y)^\top W dx,$$

so the gradient is $W(x - y)$. If $W \succ 0$, this quadratic is the nonconstant part of a Gaussian negative log-likelihood. If W is singular, write $W = V_r \Lambda_r V_r^\top$ with $\Lambda_r \succ 0$; the finite observation coordinate $\Lambda_r^{1/2} V_r^\top y$ has mean $\Lambda_r^{1/2} V_r^\top x$. Whether the factor is a likelihood or a generalized-Bayes loss still depends on how W was constructed, and no projector property follows from positive semidefiniteness alone. $\square$

A.6 Fixed-energy score identity and plug-in approximation

Let $p_t(x_t) = \int q_t(x_t | x_0) p_X(x_0) dx_0$. Differentiation under the integral gives

$$\nabla_{x_t} \log p_t(x_t) = \mathbb{E} \left[-\frac{x_t - \sqrt{\bar{\alpha}_t} X_0}{1 - \bar{\alpha}_t} \middle| X_t = x_t \right].$$

Rearrangement proves the Tweedie expression in Equation (10); differentiation gives

$$J_{\hat{x}_0} = \bar{\alpha}_t^{-1/2} [I + (1 - \bar{\alpha}_t) J_{s_\theta}].$$

For any positive integrable fixed factor $L_\rho(y; x_0)$,

$$\begin{aligned} \pi_{\rho,t}(x_t | y) &= \frac{1}{Z_\rho(y)} \int p_X(x_0) q_t(x_t | x_0) L_\rho(y; x_0) dx_0 \\ &= \frac{p_t(x_t)}{Z_\rho(y)} \mathbb{E}[L_\rho(y; X_0) | X_t = x_t]. \end{aligned}$$

Taking a log gradient proves Equations (21)–(22). The identity is ordinary Bayes' rule when L_ρ is the external-mask likelihood and an algebraic Gibbs-target identity when it is an EEG-only compatibility factor.

Replacing the conditional expectation by $L_\rho(y; \hat{x}_0)$ yields

$$\nabla_{x_t} \log L_\rho(y; \hat{x}_0) = -J_{\hat{x}_0}^\top \nabla_{x_0} \mathcal{E}_\rho(\hat{x}_0; y),$$

which proves Equation (23). Expectation and evaluation at a conditional mean do not commute in general, proving Equation (43).

Before any guidance-vector normalization or clipping,

$$p_X(x) e^{-\mathcal{E}_\rho(x; y)} = \{p_X(x) e^{-\mathcal{E}_0(x; y)}\}^{1-\rho} \{p_X(x) e^{-\mathcal{E}_C(x; y)}\}^\rho,$$

because the prior exponents sum to one. Likewise, at fixed t ,

$$\nabla_{x_t} \log \tilde{q}_{\rho,t} = s_\theta(x_t, t) - \gamma_t J_{\hat{x}_0}^\top \nabla \mathcal{E}_\rho(\hat{x}_0; y),$$

for the unnormalized surrogate in Equation (44). A nonlinear map $\mathcal{T}_t$ applied only to the correction generally destroys this gradient representation. Thus the geometric identity is deliberately limited to the raw, unnormalized, unclipped

construction. Separately forward-noising the population and context clean targets also does not generally commute with geometric interpolation.

A.7 Population-posterior fallback and bounded influence

PROOF OF THEOREM 5.3. Under the stated short circuit, both NULL and direct population inference invoke the same population-posterior routine with identical arguments before any context state or context random operation is evaluated. They therefore share x_K . If the trajectories agree at reverse step k , the same state, population energy, population-guided score implementation, and random variable ζ_k are passed to the same update F_k , giving the same x_{k-1} . Backward induction proves pathwise equality and hence equality in distribution. The proof would not apply to a nominal zero multiplier placed after context-dependent normalization, clipping, or randomness. $\square$

To prove Equation (45), energy interpolation first gives $g_{\rho,t}^{\text{raw}} - g_{0,t}^{\text{raw}} = \rho(g_{C,t}^{\text{raw}} - g_{0,t}^{\text{raw}})$. Lipschitz continuity of $\mathcal{T}_t$ bounds the induced vector-field perturbation by $\rho L_{\mathcal{T}}(t)G(t)$. For $e(t) = \|x_{\rho}(t) - x_0(t)\|_2$, the upper-right derivative satisfies

$$D^+e(t) \leq L(t)e(t) + \rho L_{\mathcal{T}}(t)G(t).$$

Gronwall's integral inequality with $e(t_0) = 0$ gives Equation (45).

A.8 Bayes risk with the same observed query

Assuming the relevant conditional entropies are finite, any predictive density $q(x | y, K)$ satisfies

$$\mathbb{E}[-\log q(X | Y, K)] = H(X | Y, K) \tag{53}$$

$$+ \mathbb{E}_{Y,K} \text{KL}(p(\cdot | Y, K) \| q(\cdot | Y, K)). \tag{54}$$

The optimum with calibration summary K is therefore $H(X | Y, K)$, while the optimum that observes the same Y but not K is $H(X | Y)$. Their null-minus-calibrated difference is $I(X; K | Y)$, proving Equation (46). This result concerns the true optimal conditionals; it does not identify either learned energy as true.

For squared loss, let $\mu_K = \mathbb{E}[X | Y, K]$ and $\mu_0 = \mathbb{E}[X | Y]$. Conditional orthogonality gives

$$\mathbb{E} \|X - \mu_0\|_F^2 = \mathbb{E} \|X - \mu_K\|_F^2 + \mathbb{E} \|\mu_K - \mu_0\|_F^2,$$

because $\mathbb{E} \langle X - \mu_K, \mu_K - \mu_0 \rangle_F = 0$. Rearrangement proves Equation (47). If $K' \perp\!\!\!\perp X | Y$, conditional factorization gives $p(X | Y, K') = p(X | Y)$, so an ideal shuffled-calibration procedure equals null. There is no corresponding universal order for a plausible but mismatched operator.

More generally, let Z contain every input available to a deterministic estimator $D(Z)$, let $\mu(Z) = \mathbb{E}[X | Z]$, and let $\Sigma(Z) = \text{Cov}(\text{vec } X | Z)$. Expanding $X - D = (X - \mu) + (\mu - D)$ gives

$$\mathbb{E} [\|X - D(Z)\|_F^2 | Z] = \text{tr } \Sigma(Z) + \|\mu(Z) - D(Z)\|_F^2. \tag{55}$$

Thus the conditional mean is the Bayes action for squared error; no exact diffusion procedure can have a theorem of lower MSE than that action with the same information.

As an ideal reference only, suppose X and $X^{(1)}, \dots, X^{(K)}$ are conditionally independent draws from the same exact posterior with mean μ and covariance Σ . Covariance addition gives

$$\mathbb{E}[\|X - X^{(1)}\|_F^2 | Z] = 2 \operatorname{tr} \Sigma, \quad (56)$$

$$\mathbb{E}[\|X - K^{-1} \sum_k X^{(k)}\|_F^2 | Z] = (1 + K^{-1}) \operatorname{tr} \Sigma. \quad (57)$$

The implemented normalized and clipped trajectories have not been shown to satisfy this premise. They are therefore reported as stochastic reconstructions, and their average is a specified decision rule rather than a certified posterior mean.

Similarly, if an exact joint credible set $\Gamma_{1-\alpha}(Z)$ satisfies $\mathbb{P}(X \in \Gamma_{1-\alpha}(Z) | Z) = 1 - \alpha$, then the tower property gives

$$\mathbb{P}(X \in \Gamma_{1-\alpha}(Z)) = \mathbb{E}[\mathbb{P}(X \in \Gamma_{1-\alpha}(Z) | Z)] = 1 - \alpha.$$

This is an ideal calibration benchmark, not a property established for empirical bands from the implemented reverse process. Coordinate-wise quantiles also do not imply simultaneous waveform coverage. Consequently, the experiments use neutral terms such as empirical quantile band until held-out coverage and width justify a stronger interpretation.

B Calibration, Restoration, and Rollback Algorithms

Algorithms 1–3 make the separation among population observation conditioning, support-only operator estimation, deferred single-family activation, and query inference operational. Every preprocessing rule, retained regressor rule, rank, threshold, amendment, attenuation mapping, energy normalization, clipping operation, and reverse schedule is frozen before its designated untouched outer split is opened.

The support–query distance can be evaluated after the first query buffer arrives, but it may only lower ρ_ω or trigger NULL; it may not update V_R , $\widehat{C}_{\lambda,R}$, or $\widehat{U}_r$ from the query.

The initial manuscript protocol is therefore P0-or-NULL. B1–B6 remain deferred rather than forming an automatic fallback ladder. A backup becomes confirmatory only after the P0 mechanism has passed the prespecified RQ1 and RQ2 gates, one failure diagnostic and one family have been frozen in a prospective amendment, and evaluation moves to a new untouched outer split. The protocol never evaluates several backups on that split and retains the best. After query processing begins, monitoring may lower reliability, enter NULL, or request new support; it cannot change families within a reverse trajectory.

The population routine uses the same query, attenuation mode, population energy $\mathcal{E}_0$, normalization and clipping rules, reverse schedule, and random-stream convention as the context-capable routine. Constructing $\mathcal{W}_C$ and later multiplying its contribution by zero is deliberately disallowed. If rejection occurs after context-guided updates, restarting the complete reverse process is necessary for the pathwise equality in Theorem 5.3.

C Reproducibility Specification

C.1 Status and admission rule

This appendix is a specification for the clean-room empirical package. It does not assert that an experiment has been run or that a dataset has been acquired. The current artifact status is [TBD: review-accessible repository status and audit date]. An empirical value is admissible to the manuscript only when a provenance record identifies the dataset version and checksum, immutable split-manifest checksum, preprocessing and experiment configurations, source commit, environment or container digest, model checkpoint, all random seeds, exact command, raw output checksum, and

Algorithm 1 Support-only P0 reference-transfer calibration

Require: Disjoint support $C_\omega = \{(Y_i, E_i, \ell_i)\}$; frozen preprocessing P; retained-regressor rule R_E ; ridge λ ; top-rank rule R_r ; bootstrap count K_b ; reliability calibrator g_ψ ; hard-failure rules H

Ensure: Orthonormal context basis $\widehat{U}_r$, diagnostics z_ω , context reliability ρ_ω , immutable operator identifier v_ω

- 1: Align, center, filter, and standardize support with P; concatenate Y_c and E_c
- 2: **if** H detects missing channels, invalid timing, inadequate movement coverage, or montage/reference mismatch **then**
- 3: **return** $\emptyset, z_\omega, 0, \text{INVALID}$
- 4: $V_R \leftarrow R_E(E_c); \widetilde{E}_c \leftarrow V_R^\top E_c; G_R \leftarrow \widetilde{E}_c \widetilde{E}_c^\top$
- 5: **if** G_R is not positive definite on the retained space or fails the frozen excitation threshold **then**
- 6: **return** $\emptyset, z_\omega, 0, \text{UNIDENTIFIED}$
- 7: Solve $\widehat{C}_{\lambda,R}(G_R + \lambda I) = Y_c \widetilde{E}_c^\top$ without forming an inverse
- 8: $(\widehat{U}, \widehat{\Sigma}, \widehat{V}) \leftarrow \text{SVD}(\widehat{C}_{\lambda,R})$
- 9: $r \leftarrow R_r(\widehat{\Sigma}); \widehat{U}_r \leftarrow \widehat{U}_{:,1:r}; \widehat{\Pi}_C \leftarrow \widehat{U}_r \widehat{U}_r^\top$
- 10: **for** $b = 1, \dots, K_b$ **do**
- 11: Resample contiguous support blocks; repeat retained-space fitting and top- r truncation to obtain $\widehat{\Pi}_C^{(b)}$
- 12: Compute singular gap, bootstrap angles, G_R conditioning, effective size, movement coverage, support residuals, and allowed support-query shift features z_ω
- 13: **if** $H(z_\omega) = \text{FAIL}$ **then**
- 14: $\rho_\omega \leftarrow 0$
- 15: **else**
- 16: $\rho_\omega \leftarrow \text{clip}_{[0,1]} g_\psi(z_\omega)$
- 17: Hash the support manifest, preprocessing, retained space, rank, basis, diagnostics, and thresholds to obtain v_ω
- 18: Delete raw support from the online cleaner if the approved retention policy permits
- 19: **return** $\widehat{U}_r, z_\omega, \rho_\omega, v_\omega$

aggregation script. A manually transcribed number, an output without its configuration, or a result inherited from the superseded project is inadmissible.

Each admitted output receives a stable identifier used by the result contracts in Appendix D. The provenance ledger is append-only: rerunning an analysis creates a new record and never overwrites the command, checksum, or decision status of an earlier result. The final PDF is built from machine-readable result files rather than hand-edited table cells, and a preflight check fails while any submission-facing [TBD: field] remains.

C.2 Dataset, authorization, and integrity ledger

Every resource is registered before preprocessing. The registry records its canonical citation and landing page, version or retrieval date, custodian, access route, license or data-use agreement, redistribution constraints, original ethics and consent statement, local secondary-use determination, raw file inventory, and a SHA-256 checksum for each immutable file. When redistribution is prohibited, the package publishes a checksum manifest and acquisition script or documented access procedure rather than the data. Compressed archives and extracted files have separate hashes so another group can distinguish a changed archive from a changed extraction tool.

The ingestion report records the published and local counts of participants, sessions, trials, channels, samples, annotations, and missing files as [TBD: dataset flow values]. Exclusion rules are executed by code and produce a record-level reason, never a silent deletion. Channel names are retained, and every within-dataset rename, drop, or unavailable channel is recorded. The primary analysis does not silently interpolate all resources to one montage: each

Algorithm 2 Deferred single-family activation protocol

Require: Disjoint support C_ω ; P0 fitter/check (F_0, H_0) and frozen failure classifier C_0 ; terminal route NULL; optional prospective amendment A_{pre} ; frozen deferred library $\mathcal{J}_B = \{B_1, \dots, B_6\}$; untouched outer-split identifier u

Ensure: P0, NULL, or one prospectively activated backup family; context state; diagnostics; reliability ρ ; complete trace a_ω

- 1: Fit ($\Omega_{C,0}, z_0$) $\leftarrow F_0(C_\omega)$; append configuration, diagnostics, cost, and status to a_ω
- 2: **if** $H_0(z_0) = \text{PASS}$ **then**
- 3: $\rho \leftarrow \text{clip}_{[0,1]} g_{\psi,0}(z_0)$
- 4: **return** P0, $\Omega_{C,0}, z_0, \rho, a_\omega$
- 5: $c_0 \leftarrow C_0(z_0, \text{support metadata})$; record the frozen P0 failure code in a_ω
- 6: **if** A_{pre} is absent **then**
- 7: **return** NULL, $\emptyset, z_0, 0, a_\omega$ ▷ main experiment is P0-or-NUL
- 8: **if** RQ1 oracle/restoration or RQ2 operator-specificity gate was not passed before A_{pre} was filed **then**
- 9: **return** NULL, $\emptyset, z_0, 0, a_\omega$
- 10: Read from A_{pre} one family $j \in \mathcal{J}_B$, one triggering code, its fitter/check, thresholds, estimand, and a new outer split u_{pre}
- 11: **if** c_0 is not the amended trigger, $u \neq u_{\text{pre}}$, or u was inspected when the amendment was filed **then**
- 12: **return** NULL, $\emptyset, z_0, 0, a_\omega$
- 13: **if** the family diagnostic was not frozen on earlier development contexts or required sensors/metadata are unavailable **then**
- 14: **return** NULL, $\emptyset, z_0, 0, a_\omega$
- 15: Fit only ($\Omega_{C,j}, z_j$) $\leftarrow F_j(C_\omega)$; append the single authorized attempt to a_ω
- 16: **if** $H_j(z_j) = \text{FAIL}$ **then**
- 17: **return** NULL, $\emptyset, z_j, 0, a_\omega$
- 18: $\rho \leftarrow \text{clip}_{[0,1]} g_{\psi,j}(z_j)$; hash amendment, untouched split, support, state, diagnostics, and trace
- 19: **return** $j, \Omega_{C,j}, z_j, \rho, a_\omega$

Table 4. Dataset and authorization ledger. Every field is unresolved until checked against the data custodian and the local file inventory.

Resource	Version/access record	License or DUA	Ethics/consent and secondary use	Inventory/hash manifest
Klados–Bamidis semi-simulated EEG/EOG	[TBD: landing page, version, retrieval date]	[TBD: verified terms]	[TBD: source statement and local determination]	[TBD: file count and SHA-256 manifest]
SGEYESUB-associated recordings	[TBD: custodian and access record]	[TBD: permission and constraints]	[TBD: approval, consent, and secondary use]	[TBD: inventory and SHA-256 manifest]
Eye-BCI multimodal data	[TBD: Synapse version and retrieval date]	[TBD: verified terms]	[TBD: source statement and local determination]	[TBD: file count and SHA-256 manifest]
EEGdenoiseNet	[TBD: repository release and retrieval date]	[TBD: verified terms]	[TBD: source statement and local determination]	[TBD: file count and SHA-256 manifest]

Algorithm 3 Reliability-interpolated restoration of one query window

Require: Query Y ; frozen population prior score s_θ ; population observation state Ω_0 and routine PopulationPosterior; selected family/state $(j^\star, \Omega_C^\star)$ or NULL; reliability ρ ; attenuation source A equal to an external trace or frozen EEG estimator h_ϕ ; precision builders $W_0, W_{j^\star}$; schedule γ_k ; normalization/clipping $\mathcal{T}_k$; reverse update F_k ; initial state x_K and random stream $\{\zeta_k\}$

Ensure: Registered point estimate, restoration draws, nominal predictive summary, and audit record

- 1: **if** $j^\star = \text{NULL}$ or $\rho = 0$ **then**
- 2: **return** PopulationPosterior($Y, \Omega_0, s_\theta, A, \gamma, \mathcal{T}, F, x_K, \{\zeta_k\}$) $\triangleright$ short-circuit before constructing or applying a context operator
- 3: $y \leftarrow \text{vec}(Y)$
- 4: Obtain $a = (a_\tau) \leftarrow A(Y)$, where an external trace ignores the argument Y ; record EXTERNAL-LIKELIHOOD or EEG-GENERALIZED-BAYES mode
- 5: $\mathcal{W}_0 \leftarrow W_0(\Omega_0, a)$; $\mathcal{W}_C \leftarrow W_{j^\star}(\Omega_C^\star, a)$
- 6: Verify $\mathcal{W}_0 = \mathcal{W}_0^\top \succeq 0$ and $\mathcal{W}_C = \mathcal{W}_C^\top \succeq 0$
- 7: $\mathcal{W}_\rho \leftarrow (1 - \rho)\mathcal{W}_0 + \rho\mathcal{W}_C$
- 8: Save x_K , random-stream state, ρ , attenuation digest, family, selector trace, and population/context operator identifiers
- 9: **for** $k = K, K - 1, \dots, 1$ **do**
- 10: $s_0 \leftarrow s_\theta(x_k, k, c_g)$; $\hat{x}_0 \leftarrow [x_k + (1 - \bar{\alpha}_k)s_0] / \sqrt{\bar{\alpha}_k}$
- 11: $r_k \leftarrow \mathcal{W}_\rho(\hat{x}_0 - y)$ using a matrix-free or factored application
- 12: $g_k^{\text{raw}} \leftarrow -\gamma_k J_{\hat{x}_0}(x_k, k)^\top r_k$ using a vector-Jacobian product
- 13: $s_k^{\text{impl}} \leftarrow s_0 + \mathcal{T}_k(g_k^{\text{raw}})$
- 14: $x_{k-1} \leftarrow F_k(x_k, s_k^{\text{impl}}, \zeta_k)$
- 15: Repeat with preregistered streams for restoration draws, or compute the registered deterministic decision rule
- 16: Evaluate monitor features and append them to the context audit log
- 17: **if** the window monitor rejects calibration **then**
- 18: Restore the saved x_K and random stream
- 19: **return** PopulationPosterior($Y, \Omega_0, s_\theta, A, \gamma, \mathcal{T}, F, x_K, \{\zeta_k\}$) $\triangleright$ explicit full population-posterior restart
- 20: **if** rejection persists for the frozen change-point horizon **then**
- 21: Invalidate v_ω and request new support; remain on PopulationPosterior until calibration passes
- 22: **return** the registered point estimate, restoration draws, nominal predictive summary, and audit record

admitted dataset-montage-reference configuration has a separately keyed clean prior and population observation model. Any file whose checksum, dimensionality, sampling rate, event timing, or declared participant/session identity fails validation is quarantined until the discrepancy is resolved.

C.3 Immutable context and support-query manifests

The outer manifest is generated once from raw-record identifiers and committed before model development. Its row key contains dataset, participant, session, montage, electrode-state label when available, recording, raw time block, and event or trial identifier. It assigns each row to population training, training-context validation, held-out support, held-out query, or excluded. A participant identifier may occur in only one outer group. In session-transfer experiments, all sessions of the held-out participant remain absent from population training, and the manifest explicitly identifies which session supplies support and which supplies query.

The support manifest expands every held-out context over the common grid 0, 5, 10, 30, 60, and 120 seconds subject to a dataset-specific availability flag. It stores the selection rule, support-selection seed, exact raw intervals, regressor-excitation coverage, guard intervals, paired query identifiers, intervention assignment, P0 eligibility and failure codes, reliability version, population-operator identifier, and the sum of unique native support durations. Because one native Klados–Bamidis recording contains only 30 seconds, its 60- and 120-second entries are valid only when a preregistered concatenation draws distinct, query-disjoint calibration blocks from multiple recordings of the same held-out context. Repetition, overlapping extraction, or time resampling cannot increase the support budget; an unsupported row is explicitly marked unavailable. A zero-second row has no support interval, sets the context gate to zero, and retains the same-query population observation energy. Correct support shares the query context but no raw sample or event; wrong support carries a different context identifier; shuffled support records both the permutation seed and its stratum; the population condition resolves only to training contexts; and oracle support records the data-generating operator identifier without exposing it to model development. Attempted-family order and routing fields are added only in a prospectively amended deferred-portfolio experiment.

Automated leakage tests assert participant and session grouping, support/query interval disjointness, non-overlapping filter guard regions, absence of duplicated source segments across mixture folds, train-only normalization and threshold fitting, and the absence of test identifiers from configuration-selection logs. The manifest hash is embedded in every checkpoint and output. A changed split therefore invalidates rather than silently reuses downstream artifacts. The final hashes are [TBD: outer manifest SHA-256] and [TBD: support/intervention manifest SHA-256].

C.4 Preprocessing parity

All methods begin from the same immutable channel-mapped signal for a given deployment condition. Raw blocks are assigned before preprocessing and are processed independently across support/query guard boundaries. The machine-readable preprocessing configuration records referencing, units, polarity, bad-channel rule, causal or zero-phase filter implementation, passbands and notches, filter order and padding, resampling algorithm and target rate, synchronization correction, event alignment, amplitude clipping policy, normalization statistic and fitting scope, window length and stride, external-event or attenuation labels, and missing-data behavior. Their frozen values remain [TBD: complete preprocessing configuration and checksum].

The executable order is raw block selection, channel mapping, referencing, filtering, resampling, alignment, normalization with population-training statistics, and segmentation. If a baseline’s published method requires a different native operation, the deviation is declared in its configuration and a parity analysis separates that operation from the cleaning comparison. No baseline receives a cleaner input, longer training exposure, test-context normalization, or a more favorable channel set without an explicitly labeled assumption-matched analysis. Window overlap is permitted only within an assigned block and never creates cross-fold or support–query duplicates. Semi-simulated clean and artifact sources are folded before mixtures are generated.

C.5 Model, calibration, decoder, and optimization records

Every executable experiment resolves a versioned configuration rather than relying on command defaults. Table 5 lists the minimum fields. Architecture records include tensor layout, channel order, receptive field, layers, widths, kernels, dilation, normalization, activation, residual and attention blocks, conditioning path, attenuation injection, parameter count, and initialization. Training records include objective terms and weights, optimizer and scheduler, learning rate, batch construction, update count, early stopping, gradient clipping, precision, exponential moving average,

checkpoint-selection rule, and training hardware. Each population package binds the clean-prior checkpoint to a dataset, montage, reference, preprocessing key, and training-only observation energy. It also states whether the neural score model itself is unconditional or observation-conditional so the population and context energies cannot be double counted.

Calibration records include the semantic identity of each EOG or eye-tracking regressor, centering and scaling, support-only EOG-to-EEG transfer fit, retained rank and training-only rank-selection rule, ridge value, singular spectrum and gap, exact top- r QR/SVD projector, bootstrap scheme, subspace distance, excitation statistic, conditioning statistic, support-query shift features, and failure thresholds. Observation records bind the training-only population energy $\mathcal{E}_0$, context energy $\mathcal{E}_C$, bridge convention, variance, and attenuation a_τ . They state whether a_τ is supplied by an external sensor, in which case the conditional likelihood interpretation is used, or predicted from EEG, in which case the computation is labeled a generalized-Bayes observation energy. The attenuation is never logged as an artifact probability unless a separate probability-to-variance model is specified. Deferred-portfolio records are dormant in the primary campaign and acquire robust, lagged, filter-bank, graph, CCA/GED, shrinkage, routing, and compute fields only under a prospective protocol amendment. Gate records include features, architecture or rule, fitting loss, calibration method, thresholds, clipping, and proof that the per-window value is fixed with respect to the diffusion state.

Diffusion records include forward schedule, score/noise/ x_0 parameterization, loss weighting, sampler, reverse steps, stochasticity, population and context guidance signs, observation variance, Jacobian or stopped-gradient choice, normalization/clipping, restoration decision rule, and number of stochastic draws. The information-matched one-step record establishes the same clean training records, synthetic partial-observation construction, available inputs, temporal receptive field, and conditioning. The strong task-matched record separately declares paired noisy-clean supervision and corruption exposure. Both report parameter and optimization-budget differences. Decoder records include trial input, architecture, target, class balance, augmentation, optimizer, selection metric, calibration, subject/session aggregation, and the distinction between fixed and retrained evaluation.

C.6 Baseline provenance and parity audit

Each external method has an upstream URL, citation, license, exact source commit or release, environment, pretrained-checkpoint provenance if used, local patch, and command. A local reimplementaion is labeled as such and is validated on an upstream example or invariant before comparison. Baselines are not silently simplified to fit a common interface. Method-specific assumptions—EOG access, calibration duration, clean reference data, montage, window size, or channel count—are reported beside results, and both native-assumption and common-information comparisons are produced when they answer different questions.

C.7 Environment, randomness, and executable output ledger

The repository records the operating system, compiler and driver, Python and accelerator runtime, exact dependency lock, container recipe and digest, CPU, GPU, memory, storage, precision, deterministic-kernel settings, and known nondeterministic operations as [TBD: environment lock and hardware inventory]. Training, support selection, mixture generation, shuffled intervention, sampler, bootstrap, and permutation tests use separate named random streams. The seed manifest contains [TBD: preregistered seed values and derivation rule]. Seeds are common across paired methods, and a failure to reproduce bitwise output is quantified rather than hidden.

Every table and figure is produced by one noninteractive command from a clean checkout. A command record states its output identifier, working directory, commit, environment digest, configuration and manifest hashes, checkpoint

Table 5. Configuration freeze and checksum contract. The values are placeholders, not recommended defaults.

Component	Required immutable fields	Selection boundary	Frozen value and checksum
Preprocessing	Channel map; reference; filters; rate; normalization; windows	Training contexts only	[TBD: configuration URI and SHA-256]
Population prior	Architecture; parameterization; objective; optimizer; checkpoint rule	Training and validation contexts only	[TBD: configuration, count, and checksum]
Population/context observations	Population bank; EOG transfer; P0 rank; $\mathcal{E}_0, \mathcal{E}_C$; variance	Training validation, then support-only context fit	[TBD: configuration and checksum]
Attenuation and gate	External/EEG-only mode; inputs; calibration; thresholds; clipping; rollback	Training and validation contexts only	[TBD: configuration and checksum]
Sampler/guidance	Schedule; steps; energy bridge; Jacobian; clipping; draws; decision rule	Training validation only	[TBD: configuration and checksum]
One-step estimators	Information-matched versus task-matched data; inputs; capacity; loss; budget	Same outer folds as diffusion	[TBD: match audit and checksum]
Downstream decoders	Inputs; architecture; target; aggregation; fixed/retrained rules	Training and validation contexts only	[TBD: configuration and checksum]

Table 6. Baseline provenance ledger. One populated machine-readable row is required per implemented variant.

Family	Upstream re-lease/license	Local modification	Data and budget parity	Validation artifact
Regression / ICA / ASR / Wiener filter	[TBD: repositories, versions, licenses]	[TBD: patch identifiers]	[TBD: inputs and fitted-data audit]	[TBD: test and checksum]
Native SGEYESUB correction	[TBD: repository or faithful specification]	[TBD: sequential correction parity]	[TBD: support and channel parity]	[TBD: test and checksum]
Orthogonal subtraction controls	[TBD: source commit and specification]	[TBD: oracle/estimated projector implementation]	[TBD: span, timing, and rank parity]	[TBD: test and checksum]
Deterministic deep denoisers	[TBD: release per model]	[TBD: patch per model]	[TBD: fold, capacity, and update audit]	[TBD: test and checksum]
Published diffusion denoisers	[TBD: release per model]	[TBD: patch per model]	[TBD: fold, sampler, and compute audit]	[TBD: test and checksum]
Population and matched internal controls	[TBD: source commit]	[TBD: variant configuration]	[TBD: backbone, input, and checkpoint audit]	[TBD: test and checksum]

hashes, seed-manifest hash, standard output/error log, wall-clock timestamps, exit status, and output SHA-256. The final command index remains [TBD: per-output command ledger]. Raw per-window outputs are retained for audit, but inferential tables are generated only from the participant/session aggregation script specified in Appendix D.

Mandatory primary-campaign tests check dimensions and vectorization order; top- r projector symmetry, rank, and idempotence; retained-regressor excitation and singular-gap failures; reliable-complement cancellation on a synthetic

transfer fixture; population and context energy gradients; positive-semidefinite bridged precision; support/query disjointness; fold and normalization leakage; absence of query-target fields from fitting code; fixed attenuation and fixed-gate gradients; guidance sign and factor; an explicit zero-gate short-circuit; pathwise equality to a direct population-posterior invocation under shared randomness; metric definitions; context-level aggregation; Holm adjustment; and the absence of unbound test-context hyperparameters. The short-circuit test also asserts that no context operator, residual, Jacobian product, normalization, or clipping function is called. Block-robust, lag, filter-bank, graph, CCA/GED, shrinkage, routing, and terminal-refusal tests remain specified but are not required to launch the primary vertical slice; they become mandatory before any prospectively activated backup experiment. Numerical tolerances and fixture checksums remain [TBD: test specification and continuous-integration record]. No empirical output is admitted when a relevant test fails.

C.8 Research-process assistance and human verification

An AI-use audit is retained because OpenAI Codex contributed to research-lifecycle tasks in the clean-room process. The audit records [TBD: tool/model version and access dates], the exact prompts or task descriptions, inputs supplied, outputs retained, and the files or decisions influenced. Retained assistance includes manuscript structuring, experiment specification, symbolic-derivation checking, literature-query support, and code or figure drafting where applicable. Prohibited uses include inventing empirical values, accepting a citation without a primary-source audit, making an authorship decision, and providing final scientific or ethical approval.

For each retained output, a named human verifier records the independent check: re-derivation for mathematics, primary-source comparison for citations, code review plus executable tests for implementation, and source-data/provenance inspection for figures and results. The completed verifier roster, dates, and sign-offs remain [TBD: human verification ledger]. The final disclosure is updated to match the ACM authorship and AI policy in force on the submission date; AI systems are not listed as authors.

C.9 Availability and archival plan

The review artifact will provide [TBD: anonymous or author-visible review-access URL consistent with journal instructions]. The publication snapshot will be archived at [TBD: permanent repository and DOI] and will include source, environment lock or container recipe, configurations, manifests that can be shared, checksum lists for restricted data, preprocessing and evaluation code, tests, checkpoint provenance, raw metric outputs where permitted, and commands for every display item. For data or model artifacts that cannot be redistributed, the archive will state the legal or ethical reason and supply the closest usable access procedure or synthetic fixture. A mutable personal webpage or an unsupported “available after acceptance” promise does not satisfy this contract.

D Metric Definitions and Empty Result Contracts

This appendix fixes how empirical values enter the manuscript. The tables are empty by design. A value may replace a [TBD] token only when it can be regenerated from a provenance record described in Appendix C; manually copied console output and illustrative values are inadmissible.

D.1 Paired fidelity and subspace-resolved errors

For a clean target $X \in \mathbb{R}^{C \times T}$ and decision estimate $\tilde{X}$, temporal relative root-mean-square error is

$$\text{RRMSE}_t = \frac{\|\tilde{X} - X\|_F}{\|X\|_F}. \quad (58)$$

Let $\mathcal{F}$ denote the frozen discrete Fourier transform and W_f the prespecified frequency weights after removal of excluded bins. Spectral error is

$$\text{RRMSE}_f = \frac{\|W_f \odot (|\mathcal{F}\tilde{X}| - |\mathcal{F}X|)\|_F}{\|W_f \odot |\mathcal{F}X|\|_F}. \quad (59)$$

Channel-wise temporal correlation is Fisher transformed before context aggregation and transformed back only for presentation. Signal-to-noise improvement uses one frozen convention,

$$\Delta \text{SNR} = 10 \log_{10} \frac{\|X\|_F^2}{\|\tilde{X} - X\|_F^2} - 10 \log_{10} \frac{\|X\|_F^2}{\|Y - X\|_F^2}. \quad (60)$$

An explicit epsilon and the treatment of zero-energy windows are fixed before evaluation.

On semi-simulated data with oracle Π , the restoration target is isolated by

$$e_{\parallel} = \frac{\|\Pi(\tilde{X} - X)\|_F}{\|\Pi X\|_F}, \quad e_{\perp} = \frac{\|(I - \Pi)(\tilde{X} - X)\|_F}{\|(I - \Pi)X\|_F}. \quad (61)$$

The first quantity tests neural recovery where subtraction deletes information; the second checks consistency in directions declared reliable. Undefined denominators are handled by a preregistered exclusion or epsilon rule and reported, never silently replaced.

The practical size of the overlap is reported independently as

$$r_{\text{overlap}} = \frac{\|\Pi X\|_F^2}{\|X\|_F^2}. \quad (62)$$

A small $e_{\parallel}$ cannot establish practical importance when r_{overlap} is negligible. Because the observation energy has finite variance, complement error is an empirical constraint diagnostic rather than a test of an asserted equality $Q\tilde{X} = QY$.

D.2 Natural-recording attenuation and preservation

Artifact attenuation is expressed as a paired log ratio or decibel change between raw and cleaned blink-locked amplitude, EEG-EOG cross-spectral association, or eye-tracking-locked residual. Each metric names its channels, reference, event window, baseline, spectral estimator, and aggregation. A decrease is not automatically beneficial. Every attenuation endpoint is paired with a preservation endpoint evaluated on task-relevant or clean intervals.

For event-related potentials, the contract records amplitude, latency, and topographic similarity in frozen regions and time windows. For steady-state visual responses it records target-frequency amplitude, phase, and neighboring-bin noise. For motor imagery it records band-limited event-related desynchronization/synchronization and spatial covariance. Decoder utility is the within-context difference from raw or the declared baseline for both a frozen decoder and a decoder retrained under an identical nested split. Calibration, classifier threshold, area under the receiver-operating curve, balanced accuracy, or information-transfer rate is reported only when the outcome and class balance support that interpretation.

D.3 Uncertainty, reliability, and deployment metrics

For paired data, a nominal $1 - \alpha$ coordinate-wise predictive interval is evaluated by empirical coverage, mean width, and a proper interval score. Joint summaries such as band power use intervals constructed directly from the plug-in guided restoration draws, not intervals propagated by an unjustified linear approximation. Empirical draw-mean error and single-draw error are separate rows. An interval is called empirically calibrated only after held-out coverage is demonstrated; it is never called a credible interval for the implemented clipped finite-step process. Selective risk is plotted against retained coverage as the reliability or sampler-dispersion refusal threshold varies on a grid frozen from validation data.

Reliability quality includes calibration error for the gate’s training-defined benefit or failure target, area under the failure-detection curve where class balance permits, false-trigger rate, detection delay, time in population-posterior rollback, recalibration success, and regret relative to the population observation model. Exact rollback additionally records the maximum absolute state difference across a shared-seed reverse trajectory at $\rho = 0$ and asserts that no context operator, residual, Jacobian product, or context clipping function was called.

Primary-operator quality is evaluated at the context level. The record contains P0 eligibility, excitation, retained rank and gap, bootstrap stability, gate value, population rollback, false acceptance of a context operator that violates the registered preservation margin, false refusal of a beneficial context correction, and regret relative to the population and oracle conditions. A deferred backup experiment, if prospectively activated, adds candidate eligibility, attempted and selected family, deterministic failure code, cost, and terminal refusal under a new protocol; it cannot change the primary gate results.

Deployment summaries report median and tail latency, throughput, support-processing time, peak memory, model and operator storage, sampler evaluations, and the prespecified hardware/precision context. Means without tail behavior are insufficient for an online claim.

D.4 Empty confirmatory tables

D.5 Figure-data schemas

Every result figure reads one tidy machine-readable file. Required common columns are ‘dataset’, ‘dataset_version’, ‘montage_key’, ‘population_model_hash’, ‘outer_fold’, ‘participant’, ‘session’, ‘support_id’, ‘query_id’, ‘method’, ‘intervention’, ‘seed’, ‘metric’, ‘value’, ‘unit’, ‘status’, ‘config_hash’, ‘commit’, and ‘provenance_id’. The support-response figure additionally requires ‘support_seconds’; the specificity figure requires ‘operator_source’ and ‘projector_angle’; RQ1 rows require ‘overlap_ratio’; the attenuation–preservation frontier requires paired ‘attenuation’ and ‘preservation’ values from the same context; and the latency frontier requires ‘hardware’, ‘precision’, ‘batch’, ‘window_samples’, ‘channels’, ‘sampler’, and ‘function_evaluations’. Primary operator rows additionally record ‘rank’, ‘singular_gap’, ‘excitation’, ‘operator_hash’, ‘attenuation_mode’, ‘population_energy_hash’, ‘context_energy_hash’, ‘rho’, and ‘rollback’. A prospectively activated backup adds family, eligibility, failure-code, attempt, selection, and cost fields under a separate schema version. Aggregation scripts retain raw context rows and write a second file containing only plotted estimates and intervals.

Figure 5 is the first-pass layout contract. It contains no data and is excluded from any claim until the schemas above are populated.

Table 7. Primary go/no-go results. Each estimate is a participant- or session-level paired effect with uncertainty; thresholds are signed before outer-test evaluation.

Question	Primary contrast / outcome	Effect and interval	Multiplicity-adjusted evidence	Gate
RQ1 mechanism	Oracle-calibrated – population and oracle-span subtraction; $e_{ }$, $e_{\perp}$, r_{overlap}	[TBD: effect, confidence interval, contexts]	[TBD: test, adjusted p, margin]	[TBD: pass/stop]
RQ2 specificity	Correct population; wrong/shuffled controls	[TBD: effect, confidence interval, contexts]	[TBD: test, adjusted p, margin]	[TBD: pass/fail]
RQ3 diffusion	CSPD – information-matched one-step; strong task-matched comparator	[TBD: effect, confidence interval, contexts]	[TBD: decision and cost rule]	[TBD: retain/remove diffusion]
RQ4 attenuation/preservation	Learned attenuation plus physiological non-inferiority	[TBD: paired attenuation and preservation intervals]	[TBD: joint gate and adjusted evidence]	[TBD: claim scope]
RQ5 rollback	Failed context calibration – population posterior; detection and regret	[TBD: detection, regret, recovery intervals]	[TBD: numerical and operational rules]	[TBD: deployment claim]
Deferred backup	One prospectively activated family after diagnosed P0 failure	[TBD: separate amended analysis]	[TBD: separate multiplicity family]	[TBD: does not alter RQ1–RQ5]

Table 8. Support-duration and operator intervention contract. Unsupported dataset–duration combinations are marked “not available,” not imputed.

Dataset / outcome	0 s	5 s	10 s	30 s	60 s	120 s
Klados–Bamidis paired fidelity	[TBD: population posterior]	[TBD]	[TBD]	[TBD]	[TBD: NA unless valid concatenation]	[TBD: NA unless valid concatenation]
SGEYESUB-associated real attenuation	[TBD: population posterior]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Eye-BCI within-session preservation	[TBD: population posterior]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Cross-session transfer / drift	[TBD: population]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Reliability and rollback rate	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
P0 eligibility / population-route rate	[TBD: population]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Table 9. Deployment and uncertainty contract at the preregistered decision setting.

Method	Function evaluations	Latency median / tail	Peak memory	Coverage / width	Failure back
Information-matched one-step	1	[TBD]	[TBD]	[TBD: if probabilistic]	[TBD]
Strong task-matched one-step	1	[TBD]	[TBD]	[TBD: if probabilistic]	[TBD]
Population-posterior diffusion ($\rho = 0$)	[TBD]	[TBD]	[TBD]	[TBD: nominal predictive]	Population reference
CSPD calibrated	[TBD]	[TBD]	[TBD]	[TBD]	[TBD: trigger, delay, regret]
Deferred backup (if activated)	[TBD]	[TBD: all attempted fits plus inference]	[TBD]	[TBD]	[TBD: separate amended protocol]

Table 10. Deferred operator-family contract. This table remains inactive in the primary campaign and is populated only after a prospective, family-specific amendment following a diagnosed P0 failure.

Candidate / policy	Eligible texts	con-	Selected texts	con-	Operator stability	sta-	Fidelity preservation	/	Failure-aware loss	Total calibration cost
P0 INST	[TBD]		[TBD]		[TBD]		[TBD]		[TBD]	[TBD]
B1 ROBUST	[TBD]		[TBD]		[TBD]		[TBD]		[TBD]	[TBD]
B2 FIR	[TBD]		[TBD]		[TBD]		[TBD]		[TBD]	[TBD]
B3 FB-COV	[TBD]		[TBD]		[TBD]		[TBD]		[TBD]	[TBD]
B4 GRAPH	[TBD]		[TBD]		[TBD]		[TBD]		[TBD]	[TBD]
B5 COV	[TBD]		[TBD]		[TBD]		[TBD]		[TBD]	[TBD]
B6 POP	[TBD]		[TBD]		[TBD]		[TBD]		[TBD]	[TBD]
Population route	Always available in-distribution		[TBD]		Training-only operator		[TBD]		[TBD]	[TBD]
Amended family protocol	By one frozen diagnostic		One terminal state per context		[TBD]		[TBD]		[TBD: separate family criterion]	[TBD: all attempts]

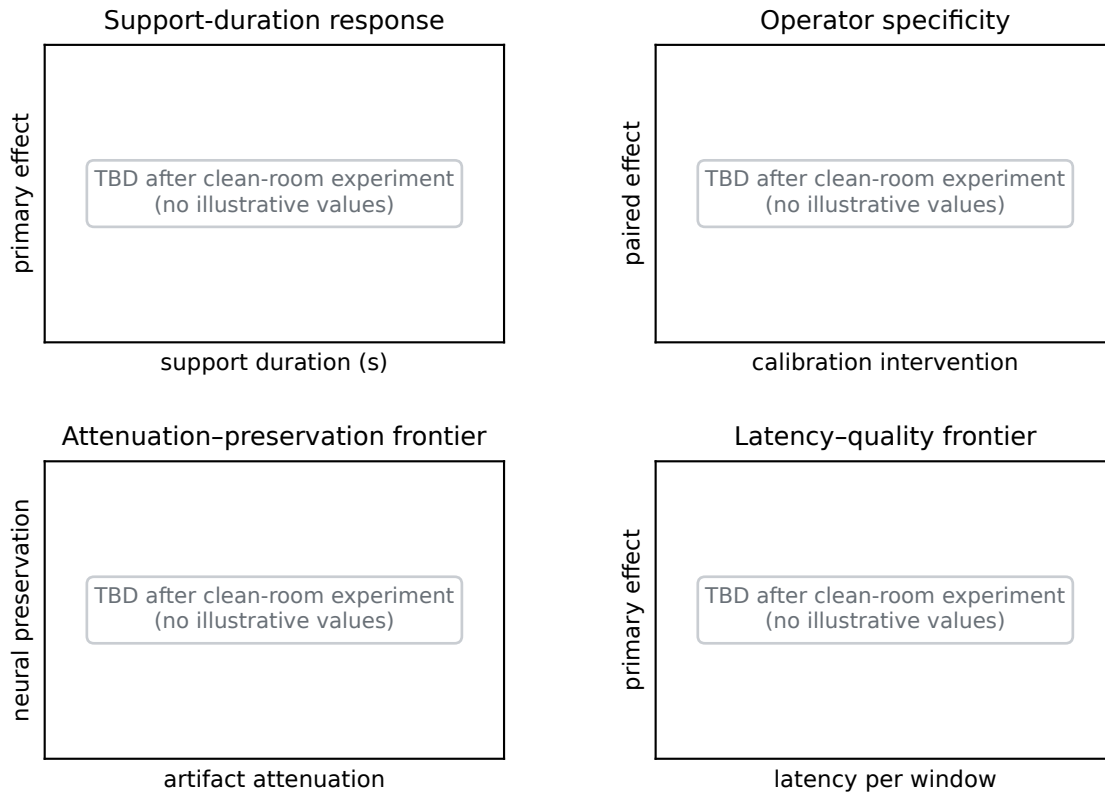


Fig. 5. Empty layouts for the registered result figures. The visible labels prevent preliminary or illustrative numbers from being mistaken for evidence. Final panels will be regenerated from the machine-readable schemas after clean-room experiments.

E Deferred Operator Contingency Portfolio

The first confirmatory campaign uses only the instantaneous ridge/SVD operator P0. The six alternatives below are retained as a design reserve for diagnosed limitations of that primary model; they are not a currently registered selector, routing ladder, or opportunity to search the present outer-test contexts. A failed primary mechanism cannot be rescued by a more elaborate operator. A backup can enter a later experiment only after P0 has passed the restoration-mechanism and operator-specificity gates, a prespecified observable failure of P0 has been documented, and a prospective amendment has frozen one family-specific protocol on a new untouched outer split.

This separation preserves the scientific question addressed by the current study. The population prior, query attenuation, population observation energy, sampler implementation, and output rule remain fixed while a future protocol changes only the support-estimated context operator. B1–B6 are therefore mathematical specifications for possible follow-up studies, not components of the current deployment policy.

E.1 A common population–context energy interface

Let $y = \text{vec}(Y)$. The training-only population state returns a fixed linear reliability map H_0 and precision $W_0 = H_0^\top H_0$. If a future protocol activates family $k \in \{P0, B1, \dots, B6\}$, its disjoint support C_ω returns a fixed context map $H_{C,k}$ and precision $W_{C,k} = H_{C,k}^\top H_{C,k}$. Both energies condition on the same query through the common interface

$$\mathcal{E}_s(x; y) = \frac{1}{2} \|H_s(x - y)\|_2^2 = \frac{1}{2} (x - y)^\top W_s (x - y), \quad s \in \{0, (C, k)\}. \quad (63)$$

For P0, and for every deferred family that returns an orthogonal projector, the context map is

$$H_{C,k} = \sigma_y^{-1} \text{blockdiag}_{\mathbb{S}_{\tau=1}}^T \left[\sqrt{w_\tau} \left(\widehat{Q}_{C,k} + a_\tau \widehat{\Pi}_{C,k} \right) \right]. \quad (64)$$

Here $a_\tau \in [0, 1]$ is the same precision-factor reliability used by the primary method: $a_\tau = 1$ retains full observation precision, whereas $a_\tau = 0$ removes precision from the selected ocular span. It is not an artifact probability. The EOG-to-EEG object estimated from calibration is consistently denoted by C_ω or an explicitly indexed estimate of it; its leading left singular subspace defines $\widehat{\Pi}_{C,k}$ when family k is projector-valued.

Context reliability ρ_ω interpolates the two energies rather than the prior and a context model:

$$\mathcal{E}_{\rho,k}(x; y) = (1 - \rho_\omega) \mathcal{E}_0(x; y) + \rho_\omega \mathcal{E}_{C,k}(x; y), \quad \nabla_x \mathcal{E}_{\rho,k}(x; y) = W_{\rho,k} (x - y), \quad (65)$$

where $W_{\rho,k} = (1 - \rho_\omega) W_0 + \rho_\omega W_{C,k}$. A rectangular or non-idempotent $H_{C,k}$ is admissible under this interface. If its precision is singular, Equation (63) is a partial likelihood on $\text{range}(H_{C,k}^\top)$; if it is positive definite, it is an anisotropic Gaussian likelihood up to a fixed normalizing constant. The fixed-energy score identity and plug-in chain rule therefore remain available, but the reliable-complement identity and projector-error bounds remain specific to exact orthogonal projectors. A covariance or filter-bank operator does not inherit those geometric claims merely by sharing the software interface.

NULL is an executable branch, not a zero context operator. It short-circuits before constructing $H_{C,k}$, a context residual, or context guidance and invokes the population-posterior procedure with $\mathcal{E}_0$, the same query, and the registered random stream. It never invokes an unconditional generator. This explicit branch is required even though the formal interpolation reduces to $\mathcal{E}_0$ at $\rho_\omega = 0$, because implementation-level normalization, clipping, and random-number consumption could otherwise violate pathwise rollback.

E.2 Deferred operator definitions

P0: instantaneous ridge/SVD transfer span. The current primary estimator is defined in Equations (12)–(13); below, $\widehat{C}_{\omega, P0} \equiv \widehat{C}_{\lambda, R}$ denotes that retained-coordinate estimate of the instantaneous transfer C_{ω} from semantically specified EOG regressors to scalp EEG. Its ridge coefficient, retained-regressor rule, rank cap, singular-gap criterion, and bootstrap diagnostics are frozen on training contexts. If P0 is invalid in the current campaign, the personalized path returns NULL; no B-family is fitted to rescue that outer-test context.

B1: block-robust EOG-transfer span. Transient electrode excursions or corrupted calibration epochs can dominate squared loss without invalidating the remaining EOG excitation. Partition support into contiguous blocks I_k and define

$$\widehat{C}_{\omega, \text{rob}} = \arg \min_D \sum_k \ell_{\mathcal{S}} \left(\frac{\|Y_{c, I_k} - DE_{c, I_k}\|_F}{s_k} \right) + \lambda \|D\|_F^2. \quad (66)$$

The robust loss, scale s_k , block length, tuning constant, optimizer, and convergence rule would be frozen by a B1-specific inner validation. A truncated SVD of $\widehat{C}_{\omega, \text{rob}}$ produces an exact projector for Equation (64). The weights may depend only on support residuals; query quality and clean targets cannot determine which blocks are down-weighted.

B2: lag-expanded EOG-transfer span. EEG and EOG can differ in alignment, phase, and waveform after acquisition filtering. Adaptive and frequency-domain regression have long been used to address such transfer effects [15, 43]. For a frozen lag set $\mathcal{L}$, B2 specifies

$$Y_c(t) = \sum_{\ell \in \mathcal{L}} C_{\omega, \ell} E_c(t - \ell) + R(t), \quad \widehat{C}_{\omega, \text{FIR}} = [\widehat{C}_{\omega, \ell}]_{\ell \in \mathcal{L}}, \quad (67)$$

with ridge and temporal-smoothness penalties chosen on training contexts. The leading left singular vectors of the concatenated transfer estimate define the candidate span. A rank cap and minimum effective-support-to-parameter ratio prevent the lag dictionary from expanding to the full sensor space. Residual coupling at preregistered nonzero lags may justify a future B2 study, but it cannot trigger an in-place family switch or be selected by improved outer-query fidelity.

B3: filter-bank anisotropic precision. A single spatial span can be inadequate when ocular transfer changes with frequency or when slow corneo-retinal activity and faster eyelid or saccadic components have different topographies. Let F_b be a frozen perfect-reconstruction or Parseval analysis filter for band b , and estimate a positive-semidefinite ocular covariance $\widehat{\Sigma}_{a, b}$ from band-limited support. Define

$$W_{b, \tau} = \left[\sigma_b^2 I + v_b(a_{\tau}) \widehat{\Sigma}_{a, b} \right]^{-1}, \quad W_{C, B3} = \sum_b F_b^{\top} \text{blockdiag}_{\tau}(W_{b, \tau}) F_b, \quad (68)$$

where the training-frozen scale $v_b(a)$ is nonincreasing, satisfies $v_b(1) = 0$, and tends to infinity as $a \downarrow 0$; $W_{b, \tau}$ at $a_{\tau} = 0$ is defined by the corresponding spectral limit, which removes precision on the retained range of $\widehat{\Sigma}_{a, b}$. Thus B3 uses the same precision-factor reliability a_{τ} as the projector families; neither a_{τ} nor $v_b(a_{\tau})$ is an artifact probability. This covariance-weighted likelihood is related in spirit to multichannel Wiener filtering [36]; it is not a projector and need not be idempotent. Filter definitions, padding, guard intervals, covariance shrinkage, the retained covariance range, and band eligibility would be frozen prospectively. The attenuation is applied in analyzed band coordinates and is not commuted through a convolution or wavelet transform without proof. If a support likelihood selects shrinkage, its fixed log-determinant term must be retained so that a diffuse covariance cannot win merely by lowering quadratic residual energy.

B4: electrode-graph-regularized EOG-transfer span. Ocular fields are spatially structured over the scalp, whereas short calibration and missing channels can produce jagged transfer estimates. Construct a montage graph from registered electrode coordinates and let $L_g \succeq 0$ be its frozen Laplacian. EEG surface-Laplacian work provides geometric motivation [3]; the following EOG-transfer estimator is a proposed contingency rather than an established correction guarantee:

$$\widehat{C}_{\omega,g} = \arg \min_D \|Y_c - DE_c\|_F^2 + \lambda \|D\|_F^2 + \eta \operatorname{tr}(D^\top L_g D). \quad (69)$$

It satisfies the Sylvester equation

$$\eta L_g \widehat{C}_{\omega,g} + \widehat{C}_{\omega,g} (E_c E_c^\top + \lambda I) = Y_c E_c^\top. \quad (70)$$

The span of $\widehat{C}_{\omega,g}$ yields an exact projector, but graph regularization can bias that span even in noiseless calibration. The span-invariance result for ordinary ridge therefore does not apply. A future B4 protocol must freeze graph construction and test graph bias, leave-electrode-out stability, coordinate reliability, and montage compatibility before opening a new outer split.

B5: contrastive covariance or CCA-GED span. When a continuous ocular reference is poorly conditioned but prompted ocular and neutral blocks are available, a covariance contrast provides a different identification route. With continuous EOG, regularized canonical correlation analysis (CCA) selects EEG filters coupled to EOG; with event-only support, a generalized eigendecomposition (GED) contrasts ocular and neutral covariance. CCA has precedent in EEG artifact separation [9], and generalized spatio-spectral covariance contrasts are established EEG filter constructions [32]; neither citation proves ocular specificity in the proposed deployment setting. In event mode, define

$$K = (\widehat{\Sigma}_q + \lambda I)^{-1/2} (\widehat{\Sigma}_o - \widehat{\Sigma}_q) (\widehat{\Sigma}_q + \lambda I)^{-1/2}, \quad P_{\text{GED}} = (\widehat{\Sigma}_q + \lambda I)^{1/2} V_+, \quad (71)$$

where V_+ contains preregistered positive, stable eigenmodes. In CCA mode, spatial filters W are converted to forward activation patterns, for example $P_{\text{CCA}} = \widehat{\Sigma}_Y W (W^\top \widehat{\Sigma}_Y W)^{-1}$, before QR/SVD construction of a projector. Unmixing filters are never treated as scalp mixing patterns. Event-only GED estimates an ocular-pattern subspace rather than the EOG-reference transfer C_ω and must be labeled accordingly. Event imbalance, task-locked neural differences, a weak eigengap, or unstable activation patterns are prospective exclusion conditions because covariance contrast can otherwise label neural activity as ocular.

B6: population-shrunk EOG-transfer span. For very short or under-excited support with a known montage and fixed regressor semantics, a training-only population transfer bank can stabilize the context estimate. After a training-frozen semantic alignment, define

$$\widehat{C}_{\omega,\text{ps}} = (1 - \eta(z_c)) \widehat{C}_{\omega,\text{P0}} + \eta(z_c) C_{\text{pop},R}, \quad \eta(z_c) \in [0, 1], \quad (72)$$

where the mapping from support diagnostics z_c to η would be calibrated on training contexts. QR/SVD produces an exact projector. The endpoint $\eta = 1$ is a population transfer and cannot be reported as personalization. A montage or regressor-semantic mismatch excludes use of the bank.

NULL and oracle controls. NULL always uses the explicit population-posterior short-circuit defined above. It is an admissible outcome, not a failed row and not an unconditional or zero-filled input. On paired semi-simulation only, the data-generating ocular span supplies an oracle ceiling after the relevant family, thresholds, and endpoints have been prospectively frozen. A post hoc best candidate on current test contexts is not an oracle and is prohibited. Robust PCA

Table 11. Deferred operator contingency portfolio. B1–B6 are mathematical designs for separate prospective studies, not routes in the current confirmatory campaign. Every future personalized family must use disjoint support and the common population–context energy interface; no family may inspect a query clean target.

Code	EEG information used	Diagnosable P0 limitation motivating a future study	Context output	Prospective exclusion examples
P0 INST	Synchronized EEG–EOG support	None; current primary model	Exact spatial projector	Poor excitation, conditioning, rank, or stability
B1 ROBUST	EEG–EOG blocks and residuals	Sparse support outliers	Exact spatial projector	Too little uncontaminated support or non-convergence
B2 FIR	Lagged EEG–EOG coupling	Delay, phase, or waveform mismatch	Exact rank-capped spatial projector	Insufficient effective samples or unstable lag span
B3 FB-COV	Bandwise EEG–EOG covariance	Frequency-dependent propagation or inadequate hard rank	Non-idempotent spatio-spectral precision	Filter leakage or unstable band covariance
B4 GRAPH	Electrode coordinates and EEG–EOG support	Noisy short-support topographies or missing channels	Exact graph-regularized span projector	Invalid montage graph or graph-sensitive estimate
B5 COV	EOG coupling or ocular/neutral event covariance	Unusable pointwise regression or absent continuous EOG	Exact activation-pattern projector	Confounded events, weak eigengap, or unstable patterns
B6 POP	Training-only montage bank and weak context support	Low-information calibration	Exact population-shrunk span projector	Montage or regressor mismatch
NULL	No accepted context operator	Invalid or unvalidated calibration	Explicit call to $\mathcal{E}_0$ population procedure	Never ineligible

or generic low-rank-plus-sparse separation remains only an exploratory stress test because both neural EEG and ocular activity can be structured, low rank, or temporally sparse.

E.3 Prospective activation and family-specific protocols

The current study does not execute an immediate P0-to-B1–B6 ladder. Before any backup is activated, P0 must first pass the primary mechanism test against population restoration and oracle-span subtraction and the specificity test against wrong-context and shuffled operators. Only then can an observable, scientifically interpretable P0 failure motivate a contingency study. Examples include support outliers, lagged residual coupling, frequency-dependent propagation, graph-stabilizable spatial instability, covariance-only calibration, or weak support with a compatible population bank. Failure of P0’s core restoration mechanism, or lack of operator specificity, terminates the personalized claim rather than authorizing a backup.

Activation requires both a prospective protocol amendment and a new untouched outer split. The amendment must name one backup family, its triggering diagnostic, support-only eligibility and refusal rules, hyperparameter grid, compute cap, comparator, primary endpoint, and success criterion before the new outer outcomes are opened. The family is then evaluated against P0 and NULL as a family-specific experiment. There is no simultaneous selector, no

deterministic priority ladder among B1–B6, and no retrospective promotion of whichever family performs best; each amendment defines only its own family-specific primary estimand. Evidence from one family cannot validate another.

All fitting and diagnostic thresholds in a future protocol must be learned within its training contexts. Query-time monitoring may reduce ρ_ω , enter the explicit population short-circuit, or request a new support episode, but it may not switch operator families in place. If a different failure appears, it requires a separate prospective amendment and another untouched evaluation split.

E.4 Fair comparison and audit record for a future family

A family-specific follow-up must give P0, the activated backup, and NULL the same native support duration, channel map, reference, preprocessing, movement annotations, population prior, query attenuation a_τ , population energy, guidance and sampler settings, stochastic seeds, query windows, and output rule. The backup may receive only the additional metadata inherent to its definition, such as registered electrode coordinates for B4 or prompted neutral blocks for event-mode B5. Candidate-specific calibration time, attempted-fit cost, memory, convergence failures, and refusal are charged to that family rather than omitted from its result.

The audit record must retain support and query identifiers, available sensors, family and version, diagnostics, eligibility decision, transfer or precision hash, rank where applicable, a_τ and ρ_ω versions, population-energy and sampler hashes, compute, and final fallback state. Outer-test clean targets, natural-recording preservation outcomes, and downstream decoder scores are excluded from fitting and eligibility inputs. Adding a family, changing its diagnostic, extending support, selecting rank from a query outcome, reporting only converged contexts, or changing the NULL branch after outcomes are known requires another prospective protocol and cannot revise the present confirmatory analysis.

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